Emercity — Full App Layout & Feature Specification

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1. Product overview

Emercity is a cross-platform household emergency app that knows who in your home is most at risk, for this hazard, at this address, right now, and tells you what to do first. It is framed for Loudoun County, Virginia, and uses real Loudoun open data where it exists, with patterns modeled on data-rich counties (Fairfax County, VA and Harris County, TX) and national sources (FEMA, NOAA, DOE, HHS, FDA) where Loudoun is thin.

This document is the complete build specification for the app only. It covers every screen, component, rule, data field, and state. It is written so it can be handed to a developer or another AI chat and built in one pass.

1.1 One-line pitch

Emercity turns a household's people, pets, medications, and address into a ranked, step-by-step plan that adapts to the hazard in front of them, and works with no signal.

1.2 The problem

- Generic preparedness templates (FEMA Make-a-Plan, Red Cross) treat every household the same. They do not know that one family member depends on an oxygen concentrator and another on refrigerated insulin.
- Official alerts tell you what is happening, not what your specific household should do about it, in what order.
- In a real emergency, people freeze, forget, and lose signal. Paper plans are forgotten in drawers.
- Loudoun County sees frequent power outages, flash flooding along creeks, winter storms, and inland effects of hurricanes. Each hazard changes who is most at risk.

1.3 Who uses it

User	Need	What Emercity gives them
Household organizer (usually a parent)	Build and maintain the family plan	Guided setup, priority order, supply tracking, drills
Other adults in the home	Know their role when something happens	Shared plan, assigned tasks, Emergency Mode on their own phone
Teens	Act if parents are away	Simple Emergency Mode, lock-screen card, offline guides
Caregiver of an elderly or medically dependent person	Keep that person safe through outages and evacuations	Outage countdown, device battery timers, critical-tier nudges
Pet owners	Not leave animals behind	Pet profiles, pet-friendly shelters, pet supplies in the go-bag

1.4 Product principles

These rules govern every design and engineering decision. When a decision is unclear, apply them in this order.

1. **Safety over cleverness.** Every recommendation must trace to a written rule and a cited source. No feature that drives an emergency action depends on AI output.
2. **Works with no signal.** Everything needed during an emergency (the plan, the priority order, the guides, the countdowns, the lock-screen card) is stored on the device and works in airplane mode.
3. **Order, not essays.** In emergencies, the app shows what to do next, in order, one step at a time. Explanations live one tap away, never in the way.
4. **Life-threatening first.** Anything that could be fatal if no action is taken is visually louder than everything else, always.
5. **Honest data.** Every live or location-based value shows its source and a "last updated" time. Stale data is labeled stale. The app never pretends to know what it does not.
6. **Private by default.** Medical details are encrypted, visible only to household members, and never sold or used for anything but the plan.

7. **Never replace 911 or official orders.** The app always defers to official evacuation orders and emergency services, and makes calling 911 one tap away in Emergency Mode.

1.5 Scope

In scope for version 1:

- Household profiles (people, pets, medications, medical devices)
- Priority engine (rule-based, ranked order with severity nudges)
- Address risk profile
- Live official alerts and nearest river gauge watch
- Outage countdown for medications, devices, and food
- Emergency Mode with hazard-specific playbooks (flood, hurricane inland effects, winter storm and outage, fire, chemical/hazmat)
- Medication barcode scan via openFDA
- Lock-screen card generator
- Practice drills
- Map: shelters, resources, meeting points
- Supply tracker and seasonal nudges
- Offline guide library, plus a subtle cloud Q&A restricted to official sources
- Shared household plan with accounts and cloud sync
- Multilingual guides and accessibility

Out of scope for version 1 (listed as future work): car-specific plans, plans for when you are away from home, school pickup coordination, neighbor check-ins, livestock, smartwatch app.

1.6 Terminology

Term	Meaning in this spec
Household	One shared plan: an address, its members, pets, and supplies
Member	A person in the household, whether or not they have the app

Term	Meaning in this spec
Account holder	A member who has signed in and can see and edit the shared plan
Dependent	A member without their own account (a child, an elderly parent)
Hazard	One of: Flood, Hurricane (inland), Winter storm, Power outage, Fire, Chemical/hazmat
Severity tier	Critical, Urgent, Important, Standard (see section 2 and section 8)
Priority order	The ranked list of people, pets, and actions for a given hazard
Playbook	The ordered set of steps Emergency Mode walks through for a hazard
Countdown	A live timer showing how long a medication, device, or food stays safe without power
Risk profile	What is known about the dangers at the household's address

2. Design system

The visual language is calm in normal use and unmistakably loud in an emergency. Two modes share one system: **Prepare** (light, calm, navy and teal) and **Emergency** (dark, high-contrast, severity colors dominate).

2.1 Color tokens

All colors are defined as tokens so the app can theme Prepare vs Emergency and light vs dark. Hex values are starting points; every text/background pair must pass WCAG AA contrast (4.5:1 for body text, 3:1 for large text and icons).

Token	Light (Prepare)	Dark (Emergency / dark mode)	Used for
`brand.primary`	#1B3A5C (navy)	#8FB8E8	App bar, primary buttons, links
`brand.accent`	#138A8A (teal)	#4FD1C5	Progress, readiness, selected chips
`surface.base`	#F7F8FA	#0E1116	Screen background
`surface.card`	#FFFFFF	#1A1F27	Cards, sheets
`text.primary`	#14181F	#F2F4F7	Body text
`text.secondary`	#5B6573	#A9B2BF	Captions, metadata, "last updated"
`border.default`	#DDE2E8	#2C333D	Card borders, dividers
`severity.critical`	#C8102E (red)	#FF4D5E	Life-threatening items
`severity.urgent`	#D95F02 (orange)	#FF9A3C	Serious harm likely without action
`severity.important`	#B38600 (amber)	#F5C542	Should be handled soon
`severity.standard`	#3D6A99 (blue-gray)	#7FA7D1	Normal tasks
`status.safe`	#1E7F4F (green)	#4ADE80	Completed, safe, checked-in
`status.stale`	#6B7280 (gray)	#9CA3AF	Data older than its freshness window
`hazard.flood`	#1F6FB2	#5AA9F0	Flood icon and banners
`hazard.winter`	#4A90A4	#8FD3E8	Winter storm
`hazard.outage`	#6B4FA0	#B49BE8	Power outage
`hazard.fire`	#B33A00	#FF7A45	Fire
`hazard.hazmat`	#7A7A00	#D4D44A	Chemical/hazmat
`hazard.hurricane`	#2F4F7F	#8AA8D8	Hurricane (inland effects)

Rules:

- Severity colors are never used for decoration. Red means "could be fatal" and nothing else.
- Color never carries meaning alone. Every severity also has an icon and a text label.
- Emergency Mode always uses the dark palette regardless of system setting, because it reads better at night and saves battery on OLED screens.

2.2 Severity tiers (visual treatment)

The severity tier is the single most important visual concept in the app. The priority engine (section 8) assigns tiers; this table defines how each one looks everywhere it appears.

Tier	Meaning	Card treatment	Icon	Motion	Sound/haptic
Critical	Could be fatal if no action is taken	Full-width card, 4 px red left bar, red tinted background (12% opacity), bold title, "CRITICAL" label pill in white-on-red	Filled triangle with exclamation	Slow pulse on the left bar (1.6 s cycle), disabled if Reduce Motion is on	One strong haptic when first shown in Emergency Mode
Urgent	Serious harm is likely without action within hours	4 px orange left bar, orange tint 8%, "URGENT" pill	Filled circle with exclamation	None	Light haptic when first shown
Important	Should be handled soon to stay safe or comfortable	3 px amber left bar, no tint, "IMPORTANT" pill outlined	Outlined circle with exclamation	None	None
Standard	Normal task	No bar, no pill	Category icon only	None	None

The user asked that the priority list show **order only**, without written reasons. The tier treatment above is how the app nudges attention toward life-threatening items without adding explanatory text. A long-press on any item reveals the reason (see section 8.7), but it is never shown by default.

2.3 Typography

- Font: system fonts (SF Pro on iOS, Roboto on Android) for legibility and Dynamic Type support. Numbers in countdowns use tabular figures so digits do not jump as they tick.
- Scale (Prepare mode, default size): Display 34/41 bold, Title 1 28/34 bold, Title 2 22/28 semibold, Headline 17/22 semibold, Body 17/24 regular, Callout 16/21, Subhead 15/20, Footnote 13/18, Caption 12/16.
- Emergency Mode bumps every size up one step and never goes below 17 pt for actionable text.
- All text supports Dynamic Type up to the largest accessibility size. Layouts must reflow (stack vertically) rather than truncate.

2.4 Spacing, shape, and layout

- 4 pt base grid; standard spacing steps 4, 8, 12, 16, 24, 32, 48.
- Screen side margin 16 pt on phones, 24 pt on tablets.
- Card corner radius 16 pt; buttons 12 pt; chips fully rounded.
- Minimum touch target 48 × 48 pt everywhere, 64 pt tall for primary actions in Emergency Mode.
- Maximum content width 640 pt on tablets (centered).

2.5 Core components

Each component is built once and reused. A developer should implement these first.

Component	Description	Variants
`AppBar`	Title, optional back, optional right action, household switcher	Prepare (navy), Emergency (black with hazard color stripe)
`TabBar`	Four tabs plus a central Emergency button (section 3)	Normal, offline (gray dot on Home), alert-active (red dot on Alerts)
`EmergencyButton`	Large circular red button in the center of the tab bar	Idle, alert-suggested (glowing ring), active (shows "Return to Emergency")

Component	Description	Variants
`PriorityCard`	One person, pet, or action in the priority order, with rank number, avatar/icon, name, tier treatment	Critical, Urgent, Important, Standard; compact and full
`CountdownTile`	Item name, live timer (HH:MM or days), progress ring, source chip	Running, paused (power restored), expired, not started
`AlertBanner`	Official alert with hazard icon, headline, issuing office, time	Warning (red), Watch (orange), Advisory (amber), Statement (blue)
`RiskChip`	One risk factor (e.g., "Flood zone AE") with level and source	High, Moderate, Low, Unknown
`StepCard`	One Emergency Mode step: big instruction, optional detail, Done/Skip buttons	Normal, critical, with timer, with checklist
`QuickAddChip`	Tappable chip in setup ("Insulin", "Dog") that adds a profile item	Unselected, selected, disabled
`SourceChip`	Small label showing data source and age ("NWS · 4 min ago")	Fresh, stale, offline cached
`MemberAvatar`	Circle with initials or photo, colored ring by top tier	Person, pet, dependent badge
`EmptyState`	Illustration, one sentence, one button	Per screen
`OfflineBar`	Thin bar at top: "Offline · showing saved plan from 2:14 PM"	Offline, reconnecting, synced
`ReadinessRing`	Circular progress showing household readiness 0–100	Small (Home), large (Plan tab)
`BottomSheet`	Modal sheet for details and edits	Half, full

2.6 Iconography

- Use one consistent outline icon set (SF Symbols on iOS, Material Symbols on Android, or a single cross-platform set such as Phosphor or Lucide).

- Each hazard has one fixed icon: water waves (flood), snowflake (winter storm), lightning bolt with slash (outage), flame (fire), hazard diamond (chemical/hazmat), spiral (hurricane).
- Members: person silhouette; pets: paw; medication: pill; device: plug; supplies: box; shelter: house with heart; hospital: cross; fire station: helmet; meeting point: flag.

2.7 Motion

- Default transitions 200–250 ms ease-out.
- Emergency Mode entry: 400 ms cross-fade to the dark palette with a brief hazard-color sweep, so the user clearly knows the mode has changed.
- Critical cards pulse slowly (section 2.2). All motion respects the system Reduce Motion setting.

2.8 Voice and tone of copy

- Prepare mode: friendly, plain, encouraging ("You're 72% ready. Two things left.").
- Emergency Mode: short imperative sentences, verb first, under 10 words where possible ("Move insulin to the cooler now.").
- Never alarmist adjectives. Severity is carried by the tier treatment, not by words like "terrifying" or "deadly."
- Reading level: grade 6 or below for all Emergency Mode text.

3. Information architecture and navigation

The app has four tabs plus a permanent center Emergency button, so Emergency Mode is always one tap away from any screen.

3.1 Tab bar

Position	Tab	Icon	Purpose
1	Home	House	Today's status: alerts, readiness, top priorities, countdowns
2	Household	People	Members, pets, medications, devices
Center	Emergency	Red circle with white exclamation	Start or return to Emergency Mode

Position	Tab	Icon	Purpose
3	Map	Map pin	Risk profile layers, shelters, resources, meeting points, river gauges
4	Plan	Clipboard	Priority orders by hazard, supplies, drills, lock-screen card, guides

Settings and the household switcher live behind the avatar in the top-right of every tab's app bar, not in the tab bar.

Tab bar behaviors:

- **Alert dot:** when an active official alert affects the household's address, a red dot appears on Home and the Emergency button gains a slow glowing ring.
- **Offline:** a gray dot on Home and the `OfflineBar` appears at the top of every screen.
- **During Emergency Mode:** the tab bar is hidden inside the Emergency flow. Leaving the flow (via "Minimize") shows the tab bar again, and the center button changes to "Return" with a pulsing red ring until Emergency Mode is ended.

3.2 Full screen map

Every screen in the app, grouped by where it lives. Screen IDs (like `HOME-01`) are used throughout this document.

Onboarding and account (section 5)

1. `ONB-01` Welcome
2. `ONB-02` Create account / sign in
3. `ONB-03` Join an existing household (invite code) or create new
4. `ONB-04` Address entry and confirmation
5. `ONB-05` Who lives here (quick-add chips)
6. `ONB-06` Member detail mini-forms (one per member added)
7. `ONB-07` Pets mini-forms
8. `ONB-08` Medications and devices (with barcode scan entry point)
9. `ONB-09` Risk profile reveal
10. `ONB-10` Your first priority order (reveal)
11. `ONB-11` Permissions (location, notifications, camera)

12. `ONB-12` Done: readiness score and next steps

Home (section 6)

1. `HOME-01` Home dashboard
2. `HOME-02` Alert detail
3. `HOME-03` All countdowns

Household (section 7)

1. `HH-01` Household overview (members, pets grid)
2. `HH-02` Member profile
3. `HH-03` Edit member
4. `HH-04` Pet profile
5. `HH-05` Medication detail
6. `HH-06` Add medication (scan or manual)
7. `HH-07` Scan barcode (camera)
8. `HH-08` Scan result confirmation
9. `HH-09` Device detail (medical equipment)
10. `HH-10` Invite member to the app

Emergency (section 12)

1. `EM-01` Hazard picker
2. `EM-02` Emergency home (active)
3. `EM-03` Step-by-step playbook
4. `EM-04` Priority order (emergency view)
5. `EM-05` Countdowns (emergency view)
6. `EM-06` Shelter and route
7. `EM-07` Family status board
8. `EM-08` End Emergency Mode / after-action

Map (section 16)

1. `MAP-01` Map home with layer toggles
2. `MAP-02` Place detail sheet (shelter, hospital, fire station)
3. `MAP-03` River gauge detail
4. `MAP-04` Meeting point editor

5. `MAP-05` Risk profile detail (also reachable from Plan)

Plan (sections 8, 9, 14, 15, 17, 18)

1. `PLAN-01` Plan home
2. `PLAN-02` Priority order by hazard
3. `PLAN-03` Address risk profile
4. `PLAN-04` Supplies (go-bag and home kit)
5. `PLAN-05` Supply item detail
6. `PLAN-06` Drills home
7. `PLAN-07` Drill in progress
8. `PLAN-08` Drill results
9. `PLAN-09` Lock-screen card builder
10. `PLAN-10` Guide library
11. `PLAN-11` Guide article
12. `PLAN-12` Ask (grounded Q&A)
13. `PLAN-13` Readiness score breakdown

Settings (section 22)

1. `SET-01` Settings home
2. `SET-02` Account and household members
3. `SET-03` Notifications
4. `SET-04` Language
5. `SET-05` Accessibility
6. `SET-06` Privacy and data
7. `SET-07` Offline data and storage
8. `SET-08` Data sources and credits
9. `SET-09` About and disclaimer

3.3 Deep links

Every screen has a deep link so notifications open directly to the right place:

- `emercity://alert/{alertId}` → `HOME-02`
- `emercity://emergency?hazard={hazard}` → `EM-02` with that hazard preselected
- `emercity://countdown/{itemId}` → `HOME-03` scrolled to that item

- `emercity://supplies/{itemId}` → `PLAN-05`
- `emercity://drill/{drillId}` → `PLAN-06`
- `emercity://invite/{code}` → `ONB-03` with code prefilled
- `emercity://gauge/{gaugeId}` → `MAP-03`

3.4 Global elements present on every screen

- App bar with household name (tap to switch households if the user belongs to more than one).
- Avatar button (top right) → Settings.
- `OfflineBar` when offline.
- Active alert banner pinned below the app bar when an alert is active, collapsible to a thin strip.
- Tab bar with Emergency button (hidden only inside the Emergency flow and during onboarding).

4. Data model

All household data lives in one shared household document, synced to the cloud and mirrored in an encrypted local database on each member's phone. Reference data (guides, medication storage rules, risk layers) is cached separately and is read-only.

4.1 Entity overview

Entity	Belongs to	Synced?	Notes
User	—	Yes (auth)	A signed-in person; can belong to several households
Household	—	Yes	The shared plan; one address
Member	Household	Yes	A person in the home; may or may not link to a User
Pet	Household	Yes	Animal in the home
Medication	Member or Pet	Yes	Includes storage requirements

Entity	Belongs to	Synced?	Notes
Device	Member	Yes	Electricity-dependent or critical medical equipment
Condition	Member	Yes	Mobility, hearing, vision, cognitive, pregnancy, infant, etc.
Contact	Household	Yes	Emergency contacts, out-of-area contact, doctors, vet
MeetingPoint	Household	Yes	Near-home and out-of-neighborhood spots
SupplyItem	Household	Yes	Go-bag and home kit items with expirations
Drill	Household	Yes	Drill sessions and results
Task	Household	Yes	Role assignments for Emergency Mode
EmergencySession	Household	Yes	An active or past Emergency Mode event
CheckIn	EmergencySession	Yes	A member's status during an emergency
RiskProfile	Household	Yes (computed)	Cached result of address risk lookups
Alert	Household	Local cache	Official alerts affecting the address
GaugeReading	Household	Local cache	Nearest river gauge observations and forecast
StorageRule	Reference	Bundled + updated	Per-product medication storage limits
Guide	Reference	Bundled + updated	Offline guide articles
Playbook	Reference	Bundled + updated	Emergency Mode step templates per hazard

Entity	Belongs to	Synced?	Notes
Place	Reference	Cached	Shelters, hospitals, fire stations, pharmacies

4.2 Field definitions

User

Field	Type	Notes
id	UUID	
email	string	Sign-in
displayName	string	
language	enum	en, es, ko, vi, zh-Hans, ar, hi, ur, am, fa (section 20)
householdIds	UUID[]	Households this user belongs to
memberIdByHousehold	map	Which Member record represents this user in each household
notificationPrefs	object	Section 23
createdAt, updatedAt	timestamp	

Household

Field	Type	Notes
id	UUID	
name	string	e.g., "The Patel Household"
address	object	line1, line2, city, state, zip, county, lat, lng, geocodeSource
homeType	enum	single-family, townhouse, apartment-low (floors 1–3), apartment-high (4+), mobile/manufactured

Field	Type	Notes
floorLevel	int	For apartments; affects flood and evacuation advice
hasBasement	bool	Flood risk modifier
heatingType	enum	electric, gas, oil, heat-pump, wood, other
hasGenerator	bool, generatorWatts int, generatorFuelHours int	Extends device countdowns
hasBatteryBackup	bool, batteryWh int	Home battery
vehicleCount	int	Evacuation capacity
vehicleSeats	int	Total seats; flags if members + carriers exceed seats
ownerUserId	UUID	Household admin
inviteCode	string	8-character code, rotatable
readinessScore	int 0–100	Computed (section 6.4)
createdAt, updatedAt	timestamp	

Member

Field	Type	Notes
id	UUID	
householdId	UUID	
linkedUserId	UUID or null	null = dependent without an account
firstName, lastName	string	
nickname	string	Shown on lock-screen card
photo	image ref	Optional, stored encrypted
birthDate	date	Age bands drive rules: infant <1, toddler 1–3, child 4–12, teen 13–17, adult 18–64, older adult 65+

Field	Type	Notes
role	enum	adult, teen, child, infant, older-adult
conditions	Condition[]	See below
medicationIds	UUID[]	
deviceIds	UUID[]	
bloodType	enum	Optional
allergies	string[]	Free text plus common chips (penicillin, peanuts, latex, bee stings)
primaryLanguage	enum	Guides and card render in this language for them
canDrive	bool	For evacuation planning
canSwim	bool	Flood risk modifier for children
needsHelpEvacuating	bool	Derived from conditions, can be overridden
doctorContactId	UUID	
notes	string	Private notes, max 500 chars

Condition (one record per condition, attached to a Member)

Field	Type	Notes
type	enum	mobility-wheelchair, mobility-walker, bedbound, hearing-loss, vision-loss, cognitive (dementia, autism, intellectual disability), pregnancy, infant-formula, diabetes-insulin, diabetes-oral, dialysis-home, dialysis-center, oxygen-dependent, ventilator, seizure-disorder, severe-allergy-epipen, heart-condition, respiratory (asthma, COPD), temperature-sensitive (MS, elderly), mental-health, other
severity	enum	mild, moderate, severe
notes	string	

Medication

Field	Type	Notes
id	UUID	
ownerType, ownerId	enum, UUID	Member or Pet
ndcProductCode	string	From barcode or search; null if manual
brandName, genericName	string	From openFDA or manual
strength, dosageForm, route	string	e.g., "100 units/mL", "injection, solution", "subcutaneous"
dose, frequency	string	User-entered ("10 units, twice daily")
isCritical	bool	Missing it could cause serious harm; auto-set for insulin, anticonvulsants, anticoagulants, transplant drugs, epinephrine, etc.; editable
needsRefrigeration	bool	From StorageRule or user
storageRuleId	UUID or null	Links to per-product limits
isOpened	bool, openedDate date	Changes the room-temperature window
quantityOnHand	number, unit	e.g., 3 pens
daysOfSupply	int	Computed from quantity and dose, or user-entered
refillDate	date	Reminder
pharmacyName, pharmacyPhone	string	
expirationDate	date	
packedInGoBag	bool	

Device (medical equipment)

Field	Type	Notes
id	UUID	

Field	Type	Notes
memberId	UUID	
type	enum	oxygen-concentrator, home-ventilator, CPAP/BiPAP, power-wheelchair, feeding-pump, IV-infusion-pump, suction-machine, home-dialysis, nebulizer, hospital-bed, apnea-monitor, insulin-pump, CGM receiver, other
lifeSustaining	bool	True for ventilator, oxygen (if continuous), feeding pump (if sole nutrition), dialysis; drives Critical tier
powerWatts	int	Optional; for generator math
hasInternalBattery	bool, batteryRuntimeMinutes int	From user or device manual
backupBatteryCount	int, backupRuntimeMinutes int	Spare batteries
backupNonElectric	string	e.g., "2 oxygen cylinders, about 5 hours each at 2 L/min" (user-entered)
supplierName, supplierPhone	string	Medical equipment supplier
registeredWithUtility	bool	Whether enrolled in the utility's medical priority program

Pet

Field	Type	Notes
id	UUID	
name	string	

Field	Type	Notes
species	enum	dog, cat, bird, small-mammal, reptile, fish, other
size	enum	small (<25 lb), medium (25–60 lb), large (>60 lb)
photo	image ref	Helps reunification
microchipId	string	
needsCarrier	bool, carrierLocation string	
medicationIds	UUID[]	
specialNeeds	string	e.g., heat-sensitive breed, elderly, anxious
vetContactId	UUID	
vaccinationRecord	file ref	Shelters often require it
foodDaysOnHand	int	

Contact: id, name, relationship, phone, altPhone, email, type (emergency, out-of-area, doctor, vet, pharmacy, utility, school, employer, medical-supplier), isOutOfArea bool, notes.

MeetingPoint: id, type (near-home, neighborhood, out-of-area), name, lat, lng, address, description ("the big oak by the mailbox"), photo, accessibleForMemberIds (computed: flags if a member with mobility needs cannot reach it).

SupplyItem: id, category (water, food, medical, pet, documents, tools, light-power, hygiene, clothing, baby, comfort), name, quantityNeeded (computed), quantityHave, unit, location (go-bag, home-kit, car), expirationDate, isTemplateItem (from generated list) vs custom, linkedMemberId or linkedPetId, lastCheckedDate.

Task: id, hazard (or "all"), assignedMemberId, title, order, linkedSubjectType/linkedSubjectId (e.g., "get Grandma", "grab Biscuit's carrier"), tier.

EmergencySession: id, householdId, hazard, startedBy (userId), startedAt, trigger (manual, alert-prompt), alertId, endedAt, stepsCompleted (stepId[] with timestamps and who), notes, outcome (safe-at-home, evacuated, sheltered, other).

CheckIn: id, sessionId, memberId, status (safe, need-help, on-my-way, no-response), location (optional, opt-in), message, timestamp, reportedBy (self or another member).

RiskProfile: householdId, computedAt, factors: RiskFactor[] (each: type, level high/moderate/low/unknown, value, source, sourceUrl, fetchedAt, explanation), hazardScores (per hazard 0–100).

Alert: id (from source), source (NWS, IPAWS, county), event (e.g., "Flash Flood Warning"), severity, urgency, certainty, headline, description, instruction, areaDesc, effective, expires, mappedHazard, affectsAddress bool.

StorageRule (reference): id, productName, ndcProductCodes[], unopenedRefrigerated (text), unopenedRoomTempDays, openedRoomTempDays, maxRoomTempF, minTempF (freeze threshold), notes, sourceName, sourceUrl, reviewedDate.

Place (reference): id, type (shelter, hospital, urgent-care, fire-station, pharmacy, police, cooling-center, warming-center), name, address, lat, lng, phone, petFriendly, adaCompliant, status (open, closed, unknown), capacity, currentPopulation, source, fetchedAt.

4.3 Privacy classification

Class	Fields	Handling
Sensitive health	conditions, medications, devices, allergies, bloodType, notes	Encrypted at rest on device and in the cloud; never in logs, analytics, push payloads, or URLs
Personal	names, photos, birth dates, contacts, address	Encrypted at rest; visible only to household account holders
Location (live)	CheckIn.location	Only shared during an active EmergencySession, only if the member opts in each time
Public reference	Places, guides, storage rules, risk layers	No restrictions

5. Onboarding and guided household setup

Onboarding takes a new user from download to a working, personalized priority order in under five minutes, using tappable chips instead of long forms. No AI is involved: every chip maps to fixed fields and rules.

A progress bar runs across the top of every onboarding screen (12 steps, shown as 5 labeled stages: Account, Home, People, Health, Ready). A "Skip for now" link appears on every optional step; skipped items become tasks on the Home screen.

5.1 ONB-01 Welcome

Layout, top to bottom:

1. Emercity logo (centered, 64 pt) and wordmark.
2. Illustration: a house with four figures (two adults, a child, a dog) inside a shield.
3. Headline: "Know who needs help first."
4. Subhead: "Build your family's emergency plan in five minutes. It works even without signal."
5. Three short value rows with icons: "Ranks who's most at risk for each emergency" · "Counts down how long meds last without power" · "Step-by-step help when it happens".
6. Primary button: "Get started".
7. Secondary link: "I have an invite code".
8. Language selector in the top-right corner (globe icon + current language), available before sign-in.

5.2 ONB-02 Create account / sign in

- Options: Continue with Apple, Continue with Google, Continue with email (email + password, or email magic link).
- Below buttons, one line: "Your health info is encrypted and only shared with people in your household." with a "How we protect it" link to a privacy sheet.
- Age gate: "Are you 13 or older?" Users under 13 cannot create accounts; they can only be added as dependents by an adult. Users 13–17 can join a household by invite but cannot create one (household owner must be 18+).
- Error states: email already in use (offer sign in), weak password (inline rule list), no network ("You need a connection to create an account. You can still explore a demo household." with "Explore demo" button that loads a sample Loudoun household

locally).

5.3 ONB-03 Create or join a household

Two large cards:

- **Create a new household** ("Set up your home's plan").
- **Join a household** ("Someone already invited you") → opens an 8-character code field with a QR scanner button. On valid code: shows household name and owner's first name, "Join The Patel Household?" with Join button. Then the user picks which existing member they are (or adds themselves as a new member) and skips directly to ONB-11.

5.4 ONB-04 Address

- Single search field: "Home address" with autocomplete (geocoding service; section 24).
- "Use my current location" button (requests location permission in context).
- After selection: a small map preview with a pin; user can drag the pin to correct placement. Below: confirmed address text and county detected ("Loudoun County, VA").
- Home type chips (single choice): House · Townhouse · Apartment (floors 1–3) · Apartment (floor 4+) · Mobile/manufactured home.
- If apartment: floor number stepper.
- Yes/No toggles: "Basement?" · "Generator?" (if yes: wattage optional) · "Home battery?"
- Heating chips: Electric · Gas · Oil · Heat pump · Wood · Other.
- Outside Loudoun: the app still works with national data (NWS, NWPS, FEMA, openFDA) and shows a one-line note: "Detailed local layers are available for Loudoun County. Elsewhere, Emercity uses national data."

5.5 ONB-05 Who lives here (quick-add chips)

This is the core of guided setup. It replaces the earlier AI "describe your household" idea with tappable chips.

Layout:

1. Headline: "Who lives here?"
2. Subhead: "Tap everyone in your home, including you. You can add details next."

3. A "You" card already present at the top (from the account), with an age band selector.
4. **People chips** (each tap adds one member; a counter badge shows how many): Adult · Older adult (65+) · Teen (13–17) · Child (4–12) · Toddler (1–3) · Baby (under 1) · Pregnant person.
5. **Pet chips**: Dog · Cat · Bird · Small animal · Reptile · Fish · Other pet.
6. A live "household strip" at the bottom shows avatars of everyone added so far ("You, Adult, Older adult, Baby, Dog"). Tapping an avatar removes it (with undo toast).
7. Primary button: "Next: add details".

Each chip adds a Member or Pet record with default role and placeholder name ("Older adult 1"), to be named on the next screen.

5.6 ONB-06 Member details (one card per person)

A horizontally swipeable stack of cards, one per member, with dots showing progress. Each card:

1. Name field (first name required; nickname optional).
2. Birth year (or exact date, optional). Pre-filled age band from the chip.
3. **Health and needs chips** (multi-select, grouped):
   - Medical: Insulin · Other diabetes meds · Oxygen · Ventilator · CPAP/BiPAP · Home dialysis · Dialysis at a center · Feeding pump · Seizures · Heart condition · Asthma/COPD · Severe allergy (EpiPen) · Takes daily critical meds
   - Mobility and senses: Wheelchair · Walker/cane · Bedbound · Hard of hearing / Deaf · Low vision / Blind
   - Other: Dementia/memory loss · Autism / sensory needs · Pregnant · Uses formula (babies) · Doesn't speak English well · Can't swim (children) · Heat or cold sensitive
4. "Can drive" toggle (adults/teens only).
5. Selecting certain chips opens a follow-up inline question:
   - Oxygen → "Continuous or only at night/with activity?" and "Concentrator, tanks, or both?"
   - Insulin → "Pens, vials, or pump?" and "Scan a box now or later?" (scan opens HH-07)
   - Ventilator, feeding pump, dialysis → "Does it have a battery? How long does it last?" (minutes/hours stepper)
   - Doesn't speak English well → language picker
6. Card footer: "Next person" button; final card shows "Next: pets" or "Next: medications".

5.7 ONB-07 Pets

One card per pet: name, species (prefilled), size chips (Small · Medium · Large), "Needs a carrier?" toggle, "Takes medication?" toggle, "Special needs" chips (Elderly · Anxious · Heat-sensitive · Needs special food), photo button ("A photo helps if you get separated").

5.8 ONB-08 Medications and devices

- A list grouped by member, showing what was implied by chips (e.g., "Grandma: Oxygen concentrator — add details"; "Dad: Insulin — add your insulin").
- Each row has two buttons: "Scan box" (camera, section 13) and "Type it".
- "Type it" opens a search field querying the bundled medication list and openFDA (when online) with autocomplete.
- For devices: battery runtime and backup runtime steppers, and "Registered with Dominion's medical priority list?" Yes / No / Not sure, with an info link.
- Skip link: "Add these later" (creates Home tasks marked Critical if the chip implied a life-sustaining need).

5.9 ONB-09 Risk profile reveal

While the user finished people screens, the app ran address lookups in the background (section 9). This screen reveals results as an animated stack of `RiskChip`s, each sliding in:

- "Flood zone: Minimal (Zone X)" or "High (Zone AE)" with FEMA source.
- "Nearest creek gauge: Goose Creek near Leesburg, 2.1 mi" (example; actual gauge chosen by distance).
- "Outage history: about N outages in your county over the last 12 months" (EAGLE-I).
- "Within 1 mile of a rail line" or "No rail line within 1 mile".
- "Nearest hospital: 4.3 mi".
- Summary line: "Your top risks: Power outages, Winter storms, Flash flooding."
- Button: "See who needs help first".

If lookups failed (offline), show "We'll finish checking your address when you're back online" and continue.

5.10 ONB-10 Your first priority order

The payoff screen. Shows the priority order for the household's top-risk hazard (usually Power outage for Loudoun):

- Hazard tabs across the top (Outage · Winter · Flood · Fire · Hazmat · Hurricane), top-risk hazard selected.
- Ranked `PriorityCard` list: e.g., 1. Grandma (Critical, red), 2. Dad (Urgent, orange, insulin), 3. Baby (Urgent), 4. Biscuit the dog (Important), 5. You (Standard), 6. Teen (Standard).
- One sentence below: "This order changes with each emergency. Tap a tab to see."
- Button: "Looks right" · secondary: "Something's off" (opens a sheet to adjust conditions; never lets the user drag-reorder Critical items below non-critical ones without a confirmation).

5.11 ONB-11 Permissions

One permission per card, asked in context with a clear benefit line. Each has "Allow" and "Not now".

Permission	Benefit line	If denied
Notifications (including Critical Alerts on iOS where approved, high-priority channel on Android)	"Get warned when an alert hits your address."	Home shows a persistent card to enable; alerts only visible in-app
Location (While Using; optional Always for alert accuracy when traveling)	"Find the nearest open shelter and check alerts where you are."	Uses home address only
Camera	"Scan medication barcodes."	Manual entry only
Contacts (optional)	"Pick emergency contacts quickly."	Manual entry

5.12 ONB-12 Done

- Large `ReadinessRing` animating to the starting score (often 35–55%).
- Headline: "Your plan is ready. You're 48% prepared."

- Top three next steps as tappable rows, ordered by tier (e.g., Critical: "Add Grandma's oxygen backup details"; Urgent: "Get a cooler and ice packs for insulin"; Important: "Pick a meeting spot").
- Two buttons: "Invite your household" (share invite link/QR) and "Go to Home".
- Background task: download offline package (guides, playbooks, map tiles for a 10-mile radius, storage rules) with a small progress chip "Saving for offline… 60%".

6. Home dashboard (HOME-01)

Home answers one question at a glance: "Is anything happening, and is my household ready?" Its layout changes depending on whether there is an active alert, a power outage, or neither.

6.1 Layout in normal conditions (no active alert)

Top to bottom, in a single scrolling column:

1. **App bar:** household name ("The Patel Household"), small weather glyph with current temperature from NWS, avatar button.
2. **Status hero card** (full width, 140 pt tall): green check icon, "All clear for 20175" style headline using the household's ZIP, subline "No alerts for your address · checked 3 min ago" with a `SourceChip` (NWS). Tapping opens a sheet listing the last 7 days of alerts that affected the address.
3. **Readiness card:** `ReadinessRing` (small, left) showing the score (e.g., 72), label "72% ready", and the single most important missing item as a row with its tier color ("Critical: Add backup oxygen details"). Tap → `PLAN-13`.
4. **Top priorities strip:** heading "If something happens" with a horizontal row of the first four `MemberAvatar`s in the household's top-risk-hazard order, each with a rank number badge and tier ring color. Tap → `PLAN-02` for that hazard.
5. **River gauge mini-card** (only if a gauge is within 10 miles): gauge name, current stage vs flood stage as a horizontal bar ("3.2 ft · flood stage 10 ft"), trend arrow, `SourceChip` (NOAA NWPS · 12 min ago). Tap → `MAP-03`.
6. **Countdown readiness card:** "If power went out now": three compact tiles showing the shortest safe times among the household's items (e.g., "Grandma's concentrator battery: 4 h", "Insulin (fridge): ~4 h, then cooler", "Freezer food: 48 h"). Tap → `HOME-03`.
7. **This week card:** up to three upcoming tasks (expiring supplies, refill dates, scheduled drill, seasonal nudge), each with a checkbox and tier marker.

8. **Guide of the week card:** one short guide matched to the season and risk profile (e.g., in October: "Winter outage: keeping meds and people warm"). Tap → `PLAN-11`.
9. Bottom padding so the last card clears the tab bar.

6.2 Layout with an active alert

When an alert affects the address, Home reorders so the alert dominates:

1. **Alert hero card** replaces the status hero. Background tinted with the hazard color; large hazard icon; event name ("Flash Flood Warning"); one-line instruction excerpt; "Until 9:45 PM"; issuing office ("NWS Sterling, VA"); `SourceChip`.
2. Directly below, a full-width red button: **"Start Emergency Mode for Flood"** (64 pt tall). Secondary text link: "View alert details" → `HOME-02`.
3. **Your priority order for this hazard:** the first five `PriorityCard`s (compact) for the mapped hazard, with Critical items pulsing.
4. The river gauge card moves up (if the hazard is flood).
5. Normal cards (readiness, this week, guide) move below a divider labeled "Everyday".

If multiple alerts are active, the hero shows the most severe (Warning > Watch > Advisory), with a "+2 more alerts" link.

6.3 Layout during a detected or reported power outage

The app cannot read the household's smart meter, so an outage is known in one of three ways: the user taps "Power is out" (always available on Home as a small chip in the status card), the phone reports it was unplugged from charging and Wi-Fi dropped at the same time (used only as a suggestion: "Did your power go out?" Yes/No), or county outage counts from the outage feed spike for the household's area (section 11.5).

Once the user confirms an outage:

1. Status hero turns purple (`hazard.outage`): "Power out since 7:42 PM · 1 h 18 m".
2. **Live countdown tiles** become the main content: every running countdown, sorted by time remaining, Critical items on top, using `CountdownTile` with live progress rings.
3. A button: "Open outage steps" (enters Emergency Mode with Power outage preselected).
4. A small "Power's back" button in the hero to stop countdowns (section 11.6 explains what happens then).

6.4 Readiness score (how it is calculated)

The readiness score (0–100) gives the household a single number to improve. It is fully rule-based and its breakdown is visible on `PLAN-13`.

Category	Weight	Full points when
Critical needs covered	30	Every Critical-tier person or item has its backup set up (device backup power listed, critical meds have 7+ days of supply, insulin has a cooler plan)
Household profile complete	10	Every member and pet has name, age band, conditions reviewed
Supplies	20	Go-bag and home kit items at or above needed quantities and not expired
Contacts	10	At least two emergency contacts, one out-of-area contact, doctors/vet/pharmacy for members with meds
Meeting points	10	Near-home and out-of-neighborhood points set and reachable by all members
Drills	10	At least one drill completed in the last 6 months
Household connected	5	Every adult has joined the shared plan on their phone
Alerts and offline	5	Notifications on and offline package downloaded in the last 30 days

Score rules:

- If any Critical-tier need is unaddressed, the overall score is capped at 69 and the ring shows a small red notch, no matter how complete everything else is. This prevents a household from feeling "ready" while a life-threatening gap exists.
- The score recalculates instantly on any change and syncs to all members.

6.5 HOME-02 Alert detail

- Header: hazard icon, event name, severity/urgency/certainty pills, effective and expiry times.

- "What this means for your household" box: the top three priority items for this hazard (order only, tier styling).
- Official text: headline, description, and instruction exactly as issued, in a collapsible section (with "Translate" button if the user's language is not English; translation is labeled machine-translated and the original is always one tap away).
- Map snippet showing the alert polygon and the household pin.
- Buttons: "Start Emergency Mode" (primary), "Share alert" (system share sheet), "Mark as seen".
- Footer: source and link to the official NWS alert page.

6.6 HOME-03 All countdowns

- Segmented control: Running · Ready (not started) · All.
- In normal conditions shows "Ready" countdowns: what each item's safe time would be if power went out now.
- Each `CountdownTile` opens a detail sheet: item, owner, rule used, source link, last time the user confirmed the item's state (opened vs unopened, battery charged), and what to do when time runs low.

7. Household tab

The Household tab is where the family's profile lives: every person, pet, medication, and device the priority engine reasons about. Any change here immediately re-ranks every priority order and recalculates countdowns and readiness.

7.1 HH-01 Household overview

Layout:

1. **App bar:** "Household", right action "+" (add person, pet, medication, device, contact).
2. **Summary row:** counts as small pills: "5 people · 1 pet · 6 medications · 2 devices".
3. **People grid:** two-column grid of member cards. Each card shows:
   - `MemberAvatar` (photo or initials) with a ring in the color of that member's highest tier across all hazards.
   - First name, age band label ("Older adult", "Baby").
   - Up to three small condition icons (oxygen tank, insulin pen, wheelchair, ear, eye).
   - Account status: a small phone icon if they have joined the app, or "Dependent".

    - An incomplete badge (amber dot) if required details are missing.
4. **Pets grid:** same style, paw icon, species and size.
5. **Medications section:** a compact list grouped by owner, each row showing name, strength, refrigeration snowflake icon if needed, days of supply, and a red "Critical" pill if `isCritical`.
6. **Devices section:** list with device icon, owner, battery runtime, backup runtime, and "Life-sustaining" pill if applicable.
7. **Contacts section:** emergency contacts, out-of-area contact (marked with a globe), doctors, vet, pharmacy, utility.
8. **Invite card** at the bottom if any adult hasn't joined: "2 adults haven't joined yet. Invite them so everyone has the plan." → `HH-10`.

7.2 HH-02 Member profile

Top section:

- Large avatar (96 pt), name, nickname, age, role.
- Language spoken, "Can drive" and "Needs help evacuating" indicators.

"Priority by emergency" strip (the most important block on this screen): a horizontal row of six small hazard tiles (Outage, Winter, Flood, Fire, Hazmat, Hurricane), each showing this member's rank number in that hazard's order and the tier color. This makes it obvious that Grandma is #1 in an outage but #3 in a fire.

Sections (each a card with "Edit"):

1. **Health and needs:** condition chips with severity, allergies, blood type.
2. **Medications:** rows linking to `HH-05`, with "Add medication" (scan or type).
3. **Devices:** rows linking to `HH-09`.
4. **Contacts:** doctor, specialist, medical equipment supplier.
5. **Tasks:** what this person is responsible for in each hazard (from the Task list), and who is responsible for them.
6. **Notes:** private notes (e.g., "Gets confused in the dark — keep a flashlight by her bed").
7. **Lock-screen card preview:** small preview of how this person appears on the card (section 14).

7.3 HH-03 Edit member

A form with the same chips as onboarding (section 5.6), plus fields for everything in the Member data model. Rules:

- Removing a life-sustaining condition or device asks for confirmation: "Remove oxygen? Emercity will stop treating Grandma as critical in outages."
- Saving shows a toast: "Priorities updated" with a "See changes" link that opens a before/after view of the top-risk hazard's order.

7.4 HH-04 Pet profile

- Photo (large), name, species, size, microchip number (copyable).
- Carrier: needed yes/no and where it's kept ("Garage shelf").
- Medications, food days on hand, special needs chips.
- Vet contact, vaccination record (photo or PDF upload; shown on demand at shelters).
- "Pet-friendly shelters" link → Map filtered to pet-friendly places.
- Priority by emergency strip (same as members).

7.5 HH-05 Medication detail

- Header: brand and generic name, strength, dosage form, owner.
- **Storage box:** "Needs refrigeration" (with snowflake), unopened vs opened toggle, opened date picker, and the safe-time result in plain words: "Once out of the fridge: up to 28 days below 86°F" pulled from the linked StorageRule, with `SourceChip` naming the source (e.g., FDA, manufacturer label). If no rule exists for the product: "Check the package insert or ask your pharmacist. Tap to enter the room-temperature limit from your label."
- **Supply box:** quantity on hand, dose and frequency, computed days of supply, refill date with reminder toggle, pharmacy contact with call button.
- **Critical toggle** with explanation in a tooltip: "Critical meds are ones where a missed dose could cause serious harm."
- **Go-bag toggle:** "Packed in go-bag".
- Expiration date.
- Data source line: "Product details from the FDA National Drug Code Directory" (if scanned) with the NDC.
- Delete medication (confirmation).

7.6 HH-06 Add medication

Two big buttons:

- **Scan barcode** → `HH-07` (section 13).
- **Search or type** → search field with autocomplete; results show brand, generic, strength, form. If nothing matches, "Add it manually" opens a plain form (name, strength, refrigeration yes/no, critical yes/no).

After choosing a product: owner picker (which member or pet), dose/frequency fields, opened/unopened, quantity, refill date.

7.7 HH-09 Device detail

- Device type icon and name, owner.
- **Power box:**
  - Uses electricity: yes/no.
  - Internal battery runtime (stepper in 15-minute increments).
  - Spare batteries: count and runtime each.
  - Non-electric backup (text, e.g., "2 oxygen cylinders") and estimated hours (user-entered).
  - Generator coverage: if the household has a generator and its wattage exceeds the device's watts, show "Your generator can run this" and add generator fuel hours to the countdown.
  - Total backup time (computed): "About 9 h 30 m without power".
- "Life-sustaining" toggle (auto-set for ventilator, continuous oxygen, sole-nutrition feeding pump, home dialysis).
- Medical equipment supplier contact.
- Utility medical priority registration: Yes / No / Not sure, plus an info sheet explaining that some utilities keep lists of customers who rely on medical equipment, and that being on a list does not guarantee faster restoration (so the household still needs a backup plan).
- "Battery last fully charged" date, with a monthly reminder toggle.

7.8 HH-10 Invite member

- Pick which member this invite is for (adults and teens without accounts), or "Someone new".

- Shows: 8-character invite code in large monospace text, a QR code, and buttons "Share link" (system share sheet with `emercity://invite/{code}` wrapped in a web link that opens the store if the app isn't installed) and "Copy code".
- Invite expires in 7 days; owner can revoke or regenerate.
- Teens (13–17) can be invited as members with "Teen" permissions (section 19.3).

8. Priority engine

The priority engine ranks every person and pet in the household for each hazard, assigns each one a severity tier, and outputs an ordered list. It is fully rule-based and deterministic: the same inputs always produce the same order, and every placement traces to a written rule. This is the headline feature.

Per the product decision, the list shows **order only**. No reasons are printed on the cards. Instead, the tier treatment (section 2.2) visually pushes life-threatening subjects forward: Critical subjects pulse in red, sit inside a red "Act on these first" group, and trigger a haptic the first time they appear in Emergency Mode.

8.1 Inputs

Input	Source	Example effect
Member age band	Member.birthDate	Infants and older adults rank higher in outages and fires
Conditions and severity	Member.conditions	Oxygen-dependent → Critical in outage and fire
Devices and backup power	Device	Ventilator with 2 h battery → Critical, rises within tier as battery drains
Medications	Medication	Refrigerated insulin → Urgent in outage, Critical when its countdown is under 6 h
Evacuation ability	needsHelpEvacuating, mobility conditions	Wheelchair user → Critical in fire and flash flood evacuation
Home factors	Household (floor, basement, heating, generator)	Basement bedroom → Critical in flash flood warning

Input	Source	Example effect
Hazard	Selected or mapped from alert	Switches the rule set
Live conditions	NWS forecast temperature, alert type, outage duration, time of day	Freezing temps + no heat → older adults and infants become Critical
Countdown state	Section 11	Shortest time-to-unsafe rises to the top of its tier
Pets	Pet	Ranked after humans in the same tier; capped at Important

8.2 Output

For each hazard, an ordered list of `PrioritySubject`:

- `subjectType` (member or pet), `subjectId`
- `rank` (1 = first)
- `tier` (critical, urgent, important, standard)
- `score` (0–1000, used only for ordering within a tier)
- `ruleIds[]` (which rules fired; used for the hidden long-press reason and for testing)
- `computedAt`

The engine recomputes whenever any input changes, when a new alert arrives, every 15 minutes during an active outage (countdowns change), and when the outdoor temperature forecast updates.

8.3 Algorithm

1. For the selected hazard, evaluate every tier rule (section 8.4) against each subject. A subject's tier is the **highest** tier any rule assigns it. Subjects matching no rule get Standard.
2. Compute a within-tier score (section 8.5).
3. Sort by tier (Critical → Urgent → Important → Standard), then score descending, then tie-breakers: needs help evacuating first, then youngest child, then oldest adult, then alphabetical by first name.
4. Pets: within any tier, all humans come before all pets. Pets can never exceed Important.
5. Apply user pins (section 8.8) last, subject to the Critical lock.

6. Assign ranks 1..N.

8.4 Tier rules by hazard

These are the starting rules. Each rule has an ID so it can be tested, cited, and adjusted. Thresholds are deliberately conservative: when in doubt, the rule tiers higher. Before release, have the rule set reviewed by a local emergency manager, nurse, or CERT instructor, and record the reviewer and date in the app's About screen.

Power outage (and the outage part of winter storms and hurricanes)

Rule ID	Condition	Tier
OUT-C1	Member has a life-sustaining electric device (ventilator, continuous oxygen concentrator, sole-nutrition feeding pump, home dialysis)	Critical
OUT-C2	Member's critical refrigerated medication has a countdown under 6 hours, or the medication is unsafe now	Critical
OUT-C3	Outdoor temperature at or below 32°F (forecast or current), home heat is electric or needs electricity (gas furnace blower, heat pump), no generator, and the member is an infant, older adult, or temperature-sensitive	Critical
OUT-C4	Heat index at or above 95°F and the member is an infant, older adult, pregnant, or temperature-sensitive, with no working cooling	Critical
OUT-U1	Member relies on refrigerated critical medication (insulin, some biologics) with more than 6 hours of safe time left	Urgent
OUT-U2	Member uses a non-life-sustaining electric device needed nightly (CPAP/BiPAP, power wheelchair, nebulizer, insulin pump charger)	Urgent
OUT-U3	Infant on formula that requires boiled or powered preparation, with less than 2 days of ready-to-feed formula	Urgent

Rule ID	Condition	Tier
OUT-U4	Outdoor temperature 33–45°F with no non-electric heat, and the member is an infant or older adult	Urgent
OUT-I1	Member is a child under 13, has dementia, or has autism/sensory needs (darkness and disruption)	Important
OUT-I2	Pet with refrigerated medication or heat/cold sensitivity	Important
OUT-S1	Everyone else	Standard

Winter storm applies all outage rules (assuming outage is likely) plus:

Rule ID	Condition	Tier
WIN-C1	Member requires dialysis at a center, or regular treatment that can't be skipped, and roads are forecast impassable (Winter Storm Warning or Ice Storm Warning active)	Critical
WIN-U1	Member has a heart condition (snow shoveling risk) or respiratory condition (cold air)	Urgent
WIN-U2	Member's critical medication has fewer than 5 days of supply when a Winter Storm Watch or Warning is active	Urgent
WIN-I1	Member uses a walker/cane or wheelchair (falls on ice)	Important

Flood (including flash flood)

Rule ID	Condition	Tier
FLD-C1	Flash Flood Warning or Flood Warning active AND the household is in a high-risk flood zone or within 500 ft of a stream AND the member sleeps in a basement or ground-floor room AND needs help evacuating	Critical

Rule ID	Condition	Tier
FLD-C2	Any member sleeping in a basement during a Flash Flood Warning	Critical
FLD-C3	Member uses a life-sustaining electric device and the home is in a flood zone (power is often cut during floods; evacuation may be needed)	Critical
FLD-U1	Member needs help evacuating (wheelchair, walker, bedbound, cognitive)	Urgent
FLD-U2	Child under 13 who cannot swim	Urgent
FLD-U3	Infant or toddler	Urgent
FLD-I1	Older adult without mobility needs, pregnant person	Important
FLD-I2	Any pet (pets need carriers and pet-friendly shelters)	Important

Fire (house fire and wildfire smoke/evacuation)

Rule ID	Condition	Tier
FIR-C1	Member cannot self-evacuate: bedbound, wheelchair, infant, toddler	Critical
FIR-C2	Member uses home oxygen (oxygen makes fire spread faster and hotter; equipment must be shut off and the person moved first)	Critical
FIR-C3	Member is Deaf or hard of hearing and the event happens between 9 PM and 7 AM (may not hear a smoke alarm while asleep)	Critical
FIR-C4	Member has dementia or a cognitive condition that may cause them to hide, freeze, or go back inside	Critical
FIR-U1	Child aged 4–12	Urgent
FIR-U2	Older adult, walker/cane user, low vision, pregnant person	Urgent
FIR-U3	Member with asthma/COPD (smoke)	Urgent

Rule ID	Condition	Tier
FIR-I1	Any pet (pets hide during fires; carriers needed)	Important

Chemical/hazmat (usually shelter in place; sometimes evacuation)

Rule ID	Condition	Tier
HAZ-C1	Member has severe asthma/COPD or is oxygen-dependent	Critical
HAZ-C2	Evacuation ordered AND member needs help evacuating	Critical
HAZ-U1	Infant, toddler, pregnant person, older adult	Urgent
HAZ-U2	Member with heart condition	Urgent
HAZ-I1	Anyone outdoors or away from home when the alert arrives (based on optional check-in status)	Important
HAZ-I2	Pets outdoors	Important

Hurricane (inland effects: flooding, wind, outages) evaluates flood rules, outage rules, and these wind rules, then takes the highest tier per subject:

Rule ID	Condition	Tier
HUR-C1	Mobile/manufactured home and High Wind Warning or Tornado Warning active: every member	Critical
HUR-U1	Member's critical medication or device backup lasts less than 72 hours (multi-day outages are common after hurricanes)	Urgent
HUR-I1	Member sleeps in a room with large windows on the windward side (user-marked)	Important

8.5 Within-tier score

Within each tier, the score orders subjects. Points add up to a maximum of 1000.

Factor	Points
Life-sustaining device with backup remaining under 2 h	+400
Life-sustaining device with backup remaining 2–8 h	+250
Countdown for a critical item: 400 × (1 − hoursLeft ÷ 24), floored at 0	up to +400
Cannot self-evacuate	+200
Infant	+180
Older adult 80+	+150
Older adult 65–79	+100
Each additional Critical rule fired beyond the first	+80
Each additional Urgent rule fired	+40
Cognitive condition	+60
Pregnancy	+60
Child under 13	+50

8.6 Worked example

Household: Grandma (82, continuous oxygen concentrator, battery 4 h, walker), Dad (46, insulin, opened pens), Mom (44), Baby Aria (8 months, formula), teen Rohan (15), dog Biscuit. Hazard: winter-storm power outage, outdoor temperature 24°F, gas furnace (needs electricity for the blower), no generator.

Rank	Subject	Tier	Rules fired	Why it lands here (internal)
1	Grandma	Critical	OUT-C1, OUT-C3, WIN-I1	Life-sustaining device, freezing with no heat, older adult 80+
2	Baby Aria	Critical	OUT-C3, OUT-U3	Infant in freezing home with no heat
3	Dad	Urgent	OUT-U1	Refrigerated insulin with more than 6 h left

Rank	Subject	Tier	Rules fired	Why it lands here (internal)
4	Rohan	Standard	OUT-S1	Teen, no special needs
5	Mom	Standard	OUT-S1	Adult, no special needs
6	Biscuit	Standard	OUT-S1	Pet, no special needs

The "Why it lands here" column is internal (tests and the hidden long-press reason). The user sees only ranks, names, avatars, and tier styling. If six hours pass and Dad's insulin countdown drops below 6 hours, OUT-C2 fires and Dad moves into the Critical group at rank 3.

8.7 Display (PLAN-02 and everywhere the order appears)

PLAN-02 Priority order by hazard layout:

1. Hazard tab row (scrollable): Outage · Winter · Flood · Fire · Hazmat · Hurricane. The tab for the household's top-risk hazard shows a small dot.
2. If any Critical subjects exist: a red-bordered group with the header **"Act on these first"** containing only Critical `PriorityCard`s. Cards pulse slowly. This group header is the nudge; it states priority, not reasons.
3. Below, an unlabeled list of remaining subjects in order, each `PriorityCard` showing: large rank number (24 pt, left), avatar, first name, tier pill (Urgent/Important) or nothing (Standard), and small condition icons. No reason text.
4. A thin divider and a quiet line: "Updated 2 min ago · Changes with weather and time".
5. Footer button: "Practice this order" → starts a drill for this hazard (section 15).

Hidden reasons: long-press (or VoiceOver/TalkBack custom action "Why this rank") on a card opens a small sheet: one line per rule in plain words ("Uses oxygen around the clock", "Very cold with no heat") plus sources where relevant. This is never shown by default, satisfying the order-only decision while keeping the logic inspectable.

Where the order appears: ONB-10, Home top-priorities strip, Home alert layout, HH-02 priority strip (rank per hazard), Emergency Mode EM-04, lock-screen card (top three), and printed/shared plan.

8.8 User adjustments

- Users can pin a subject higher or lower within its tier by long-press → "Move up / Move down".

- Moving a Critical subject below a non-Critical one is blocked by default; the user must confirm a sheet ("Emercity flagged Grandma as critical in an outage. Move her down anyway?"). The pin is logged and shown as a small pin icon on the card.
- Pins reset if the subject's conditions change.
- "Something's off" link on PLAN-02 opens the member edit flow, because wrong ranks almost always mean a missing or wrong condition.

8.9 Testing the engine

- Ship a test suite of at least 30 fixture households (including the worked example), each with expected orders per hazard. Every rule must be covered by at least two fixtures.
- Property tests: adding a Critical condition never moves a subject down; pets never outrank a human in the same tier; results are identical across runs.
- A debug screen (developer builds only) shows each subject's fired rules and score.

9. Address risk profile (PLAN-03 / MAP-05)

The risk profile turns the household's address into a short list of concrete risk factors, each with a level, a source, and a date, and a per-hazard risk score that decides which hazard is "top risk" across the app. It is computed once at setup, refreshed monthly, and refreshed immediately if the address changes.

9.1 Risk factors and where each comes from

Factor	How it is computed	Source (Loudoun)	Source (elsewhere / model)	Refresh
Flood zone	Point-in-polygon query of the home's coordinates against FEMA flood hazard zones	Loudoun County FEMA Flood Hazard Zones map service	FEMA National Flood Hazard Layer	Monthly
County floodplain overlay	Point-in-polygon against county major/minor floodplain	Loudoun Floodplain dataset	—	Monthly

Factor	How it is computed	Source (Loudoun)	Source (elsewhere / model)	Refresh
Distance to nearest stream / river gauge	Nearest gauge by straight-line distance	NOAA National Water Prediction Service gauges (covers Loudoun)	Same	Monthly (gauge list), live readings in section 10
Basement / ground floor	From Household fields	User	User	On edit
Outage frequency	Count of county outage events over the last 12 months and longest outage, from county-level outage history	DOE/ORNL EAGLE-I outage data	Same (national)	Yearly dataset + live feed if available
Rail line proximity	Distance from home to nearest rail line	Loudoun WebLogis rail layer	FRA national rail network	Yearly
Hazmat facility proximity	Distance to facilities that report hazardous chemicals	EPA facility registries (national)	Same	Yearly
Major highway proximity (hazmat trucks)	Distance to interstate/primary routes	County roads layer	National roads	Yearly
Distance to nearest hospital	Road or straight-line distance	Loudoun Hospitals and Urgent Care	Fairfax County hospital layer as model	Monthly
Fire station and first-due area	Which station responds first	Loudoun first dues and station locations	—	Yearly

Factor	How it is computed	Source (Loudoun)	Source (elsewhere / model)	Refresh
Home type wind vulnerability	Mobile/manufactured home flag	User	User	On edit
Heating dependence on electricity	From heating type	User	User	On edit
Regional hazard ranking	Static table from the regional hazard mitigation plan	Northern Virginia Hazard Mitigation Plan, Loudoun annex	Local plan	Yearly

Honesty rule: when a layer is modeled on another county's data or is simulated for the demo, its `SourceChip` and the Data Sources screen say so (e.g., "Shelter layer modeled on Fairfax County facility data").

9.2 Factor levels

Factor	High	Moderate	Low
Flood zone	Zones A, AE, AH, AO (1% annual chance)	Zone X shaded (0.2% annual chance)	Zone X unshaded
Distance to stream	Under 500 ft	500 ft – 0.25 mi	Over 0.25 mi
Outage frequency	Top quarter of counties nationally by events per year	Middle half	Bottom quarter
Rail line	Under 0.5 mi	0.5–1 mi	Over 1 mi
Hazmat facility	Under 0.5 mi	0.5–2 mi	Over 2 mi
Hospital distance	Over 15 mi	5–15 mi	Under 5 mi

9.3 Per-hazard risk scores

Each hazard gets a 0–100 score combining the regional base rate and address factors:

- **Power outage:** base 50 (frequent in Loudoun) + outage frequency (High +30, Moderate +15) + electric heat dependence +10 + any life-sustaining device +10.
- **Winter storm:** base 50 + electric heat dependence +15 + rural/long driveway flag (user) +10 + outage frequency High +10.
- **Flood:** base 25 + flood zone (High +45, Moderate +20) + stream distance (High +20, Moderate +10) + basement +10.
- **Fire:** base 30 (house fire applies to everyone) + oxygen in home +20 + mobility-limited member +10 + no smoke alarm confirmed (from supply checklist) +20.
- **Chemical/hazmat:** base 10 + rail (High +30, Moderate +15) + hazmat facility (High +30, Moderate +15) + highway under 0.5 mi +10.
- **Hurricane (inland):** base 25 + flood score × 0.4 + outage score × 0.3 + mobile home +20.

Scores are capped at 100. The highest score sets the "top-risk hazard" used on Home and in onboarding. The user can override the default hazard in Settings.

9.4 PLAN-03 layout

1. **Header:** address, county, small map with the household pin, "Last checked Oct 24, 2026".
2. **Your top risks:** three hazard cards ranked by score, each with hazard icon, name, a horizontal level bar (Low / Moderate / High), and one line of plain explanation ("Power outages are frequent in your county and your heat needs electricity").
3. **What we found at your address:** a list of `RiskChip`s, one per factor, with level color, value ("Zone X, unshaded"), and `SourceChip`. Tap opens a detail sheet: what the factor means, how it affects your plan, source link, data date.
4. **What this changes in your plan:** a short list of automatic adjustments, e.g., "Added sandbags and a sump pump check to your supplies", "Flood steps include moving Grandma upstairs first".
5. **Map preview button:** "See on map" → `MAP-01` with risk layers on.
6. Footer: "These are estimates from public data. Always follow official orders."

9.5 Edge cases

- Address outside every flood layer's coverage: show "Unknown" (gray) and never assume Low.
- Layer service down: use the last cached value with a stale label; if never fetched, "Couldn't check yet" with retry.
- Apartment on floor 4+: flood levels for the unit are adjusted down, but evacuation access (elevators out in outages) is flagged for members with mobility needs.

10. Live alerts and river gauge watch

Emercity watches official alert feeds and nearby river gauges for the household's address, and turns relevant alerts into a prompt to open the matching Emergency Mode. It does not read the phone's Wireless Emergency Alerts (iOS and Android do not let apps access them); it watches the same public sources those alerts come from.

10.1 Alert sources

Source	What it provides	How Emercity uses it
National Weather Service alerts API (api.weather.gov)	Warnings, watches, advisories with polygons and zone codes, in CAP format	Primary source for weather hazards
FEMA IPAWS public feed	Alerts from all authorized agencies (weather, civil, hazmat, evacuation)	Catches non-weather alerts like chemical releases or evacuation orders
County alert channels (Alert Loudoun)	County-issued notices	Not machine-readable for third parties in v1; the app links to sign-up
NOAA National Water Prediction Service API	River gauge observations, forecasts, and flood stages	Gauge watch (10.4)

10.2 How alerts reach the user

1. A cloud function polls NWS and IPAWS every 2 minutes for all zones containing at least one Emercity household.
2. For each new or updated alert, it tests whether the alert polygon (or zone, if no polygon) contains each household's coordinates.
3. If yes, it stores an Alert record for that household and sends a push notification to all household members with notifications on. The push payload contains only the alert ID and event name; no household data.
4. The app, on receiving it (or on next open), fetches full alert details, maps the event to a hazard (10.3), re-runs the priority engine for that hazard, and shows the alert UI.
5. When the app is open and online, it also polls directly every 5 minutes as a backup.

10.3 Event-to-hazard mapping and prompt behavior

Official event (examples)	Mapped hazard	Prompt behavior
Flash Flood Warning, Flood Warning	Flood	Full-screen prompt: "Flash Flood Warning for your address. Start Emergency Mode?"
Flash Flood Watch, Flood Watch, Flood Advisory	Flood	Banner + push; no full-screen prompt; suggests "Get ready" checklist
Winter Storm Warning, Ice Storm Warning, Blizzard Warning	Winter storm	Full-screen prompt
Winter Storm Watch, Winter Weather Advisory, Wind Chill Warning	Winter storm	Banner + "Get ready" checklist
Extreme Cold Warning, Extreme Heat Warning	Power outage rules (temperature)	Banner; re-ranks priorities
High Wind Warning, Severe Thunderstorm Warning	Power outage (likely)	Banner; countdown readiness card highlighted
Tornado Warning	Hurricane/wind playbook (take shelter)	Full-screen prompt with "Take shelter now" first step
Tropical Storm / Hurricane Warning (inland counties)	Hurricane	Full-screen prompt
Hazardous Materials Warning, Shelter in Place Warning, Civil Emergency Message	Chemical/hazmat	Full-screen prompt
Evacuation Immediate	Matches the alert text's hazard, else Fire/Hazmat evacuation steps	Full-screen prompt, step 1: "Leave now. Take only your go-bag."
Fire Warning, Red Flag Warning	Fire	Warning: full-screen; Red Flag: banner

The full-screen prompt (`ALERT-PROMPT`):

- Dark background, hazard icon (96 pt), event name, "Until [time]", one-line instruction excerpt.
- Primary button (red, 64 pt): "Start Emergency Mode".
- Secondary: "Not now" (keeps a red banner on Home and a glowing Emergency button).
- Tertiary link: "Read the alert".
- If the household already has an active EmergencySession for the same hazard (another member started it), the prompt says "Dad started Emergency Mode 3 min ago" and the button reads "Join".

10.4 River gauge watch

For households within 10 miles of an NWPS gauge, Emercity selects the nearest one (or up to three; user can choose which to follow on `MAP-03`).

Data shown and stored per gauge: name, ID, distance and direction from home, current stage (ft) and time observed, flood categories (Action, Minor, Moderate, Major stage in ft), forecast hydrograph points (stage over the next 48–72 hours when available), and trend (rising, falling, steady over the last 3 hours).

Gauge mini-card (Home): name, horizontal bar from streambed to Major stage with category bands colored (Action amber, Minor orange, Moderate red, Major dark red), a marker at the current stage, text "3.2 ft · 6.8 ft below flood stage", trend arrow, `SourceChip`.

MAP-03 Gauge detail:

1. Name, distance ("2.1 mi northwest of home"), source chip.
2. Large current stage number and category label.
3. Hydrograph chart: observed line (solid, past 72 h) and forecast line (dashed, next 72 h if available), with horizontal flood category bands and a "now" marker. Pinch to zoom time.
4. Category legend with the NWS meaning of each stage in plain words (Action: get ready; Minor: some roads flood; Moderate: some homes near the stream flood, evacuations possible; Major: widespread flooding).
5. Toggle: "Notify me when this gauge reaches Action stage / is forecast to reach Minor".
6. Note: "River gauges measure the stream, not your street. Small creeks and storm drains can flood faster than gauges show."

Gauge alerts:

Trigger	Notification	Tier
Forecast to reach Action stage within 24 h	"Goose Creek is forecast to reach action stage tomorrow morning." (example)	Important
Observed at Action stage	Push + Home banner	Urgent
Forecast or observed Minor flood stage or higher, and household is High flood risk	Push + full-screen prompt for Flood Emergency Mode	Critical

10.5 Alert history

- All alerts affecting the household are kept for 1 year and listed under Home → status card → "Alert history".
- Each entry: event, dates, whether Emergency Mode was started and by whom.

10.6 Failure handling

- NWS/IPAWS unreachable from the cloud: the server marks feeds degraded; app shows "Alert service delayed — check weather.gov" banner and the Home status card shows "Alerts may be delayed" instead of "All clear".
- Device offline: Home status card reads "Offline — can't check alerts. Last checked 4:12 PM." Never shows "All clear" when it cannot confirm.

11. Outage countdown

The outage countdown shows, for every power-dependent item in the home, how long it stays safe without electricity, as a live timer. It covers three kinds of items: refrigerated medications, electricity-dependent medical devices, and refrigerated/frozen food. Every number comes from a stored rule with a cited source, and every countdown carries the line "Confirm with your pharmacist or device supplier."

11.1 What gets a countdown

Item type	Starts when	Safe time based on	Ends when
Refrigerated medication (in fridge)	Outage confirmed	Fridge holding time, then product's room-temperature limit	Power restored, or moved to cooler (switches to cooler timer), or limit reached
Refrigerated medication (moved to cooler)	User taps "Moved to cooler"	Cooler holding time (user picks ice type and cooler size)	Power restored or ice exhausted
Medication at room temp (opened insulin pens, etc.)	Opened date, or outage start if user marks "removed from fridge"	Product's room-temperature days, voided above max temperature	Days limit reached
Medical device on battery	Outage confirmed	Internal battery + spare batteries + generator fuel + non-electric backup	Power restored, or user marks backup switched
Fridge food	Outage confirmed	4 hours with door closed	Power restored
Freezer food	Outage confirmed	Full freezer about 48 hours, half full about 24 hours, door closed	Power restored

11.2 Storage rules (reference data)

Medication timers use per-product `StorageRule` records, never one blanket number, because limits vary by product. Examples of how rules vary:

- Most insulin is stable at room temperature for 28 days below 86°F, but Levemir is listed at 42 days (Vanderbilt poison center).
- FDA publishes emergency guidance on insulin storage and switching products; the app's insulin rules are built from it (FDA insulin storage guidance).

- If a refrigerated item has no rule in the database, the app does not guess. It shows "No safe-time data for this product" and asks the user to enter the room-temperature limit from the package insert, labeled "from your label".

Each StorageRule stores: product name and NDC codes, unopened refrigerated guidance, unopened room-temp days, opened room-temp days, max room temperature, freeze threshold, notes (e.g., "do not freeze; do not put directly on ice"), source name, source URL, and the date the rule was reviewed. Rules ship bundled in the app and update from the server.

Food rules come from federal food-safety guidance; Loudoun County's own winter storm advice repeats the 4-hour fridge rule (Loudoun County, Jan 2026).

11.3 Countdown math

Refrigerated medication, still in the fridge:

- Phase 1 (fridge holding): 4 hours from outage start, reduced by 30 minutes each time the user logs "opened fridge".
- Phase 2 (room temperature): after phase 1, the product's room-temperature limit applies. If the indoor temperature (user-entered, or estimated from outdoor temperature when heating/cooling is out) exceeds the product's max, the countdown switches to "Unsafe — move to cooler now" immediately.
- The tile shows the phase 1 remainder prominently, because acting during phase 1 (moving to a cooler) is the most important action.

Moved to cooler: user picks cooler size (small, medium, large) and cold source (ice packs, regular ice, dry ice). Default holding times are conservative estimates shown as ranges, user-editable, with the warning "Never let insulin touch ice directly. Wrap it in a towel."

Medical device:

```
T_{backup} = T_{internal} + n_{spare} \cdot T_{spare} + T_{generator} +
T_{nonelectric}
```

where each term is the user-entered runtime and the generator term applies only if generator watts cover the device. The tile counts down from this total and shows the current stage ("Internal battery", "Spare battery 1 of 2", "Oxygen cylinder") with a button "Switched to next backup" so the user can keep the stage accurate.

Food: fixed timers, with a toggle on the freezer tile for "full" vs "half full".

11.4 Countdown tile (`CountdownTile`) layout

- Left: item icon (pill, plug, snowflake for fridge, box for freezer) inside a circular progress ring that drains as time passes; ring color follows tier (Critical red under 2 h for life-sustaining, under 6 h for critical meds).
- Center: item name ("Dad's insulin pens"), owner avatar small, current phase ("In fridge · keep door closed").
- Right: time remaining in large tabular digits ("3:12" for hours:minutes, or "26 d" for days).
- Bottom row: one action button relevant to the phase ("Moved to cooler", "Switched to spare battery", "Power's back") and a `SourceChip` ("FDA · rule reviewed Sep 2026").
- Expired state: red background, "Safe time passed", and "What to do" link to the guide (e.g., "Call your pharmacist before using").

11.5 Detecting an outage

Method	Behavior
User taps "Power is out" (Home status card, Emergency Mode, widget)	Starts all countdowns with start time = now (user can backdate: "It went out at…")
Phone heuristic: charger disconnected and home Wi-Fi lost within 60 seconds, while the phone was at the home address	Local notification: "Did your power go out?" Yes starts countdowns; No dismisses
Area outage feed shows a spike for the household's county/area	Push: "Outages reported near you. Is your power out?"
Another household member confirms an outage	All members' countdowns start, synced to the same start time

11.6 Power restored

- User taps "Power's back" (or any member does; synced).
- Each medication tile becomes a checklist: "Was the insulin kept cold the whole time?" Yes / No / Not sure. "No" or "Not sure" shows: "Call your pharmacist before using. Check for clumps, frost, or color change." plus the pharmacy call button.
- Food: a guide card on what to throw out based on elapsed time.

- Devices: "Recharge batteries now" reminder, and a 30-day reminder to check spare batteries.
- The outage is logged (start, end, duration) and contributes to the household's own outage history shown on the risk profile.

11.7 Notifications during an outage

When	Message (example)	Tier
Fridge phase ends in 30 min	"Move Dad's insulin to the cooler in the next 30 minutes."	Urgent
Life-sustaining device under 2 h	"Grandma's oxygen backup: about 2 hours left. Plan for more power or help."	Critical
Life-sustaining device under 30 min	"Under 30 minutes of backup for Grandma's oxygen. Call 911 if you have no other source."	Critical
Freezer near limit	"Freezer food: about 4 hours left if the door stays closed."	Important
Indoor temp too high for a medication	"Too warm for Aria's medicine. Move it to the cooler."	Urgent

All countdown timers run locally on the device and continue in the background via scheduled local notifications, so they work with no signal.

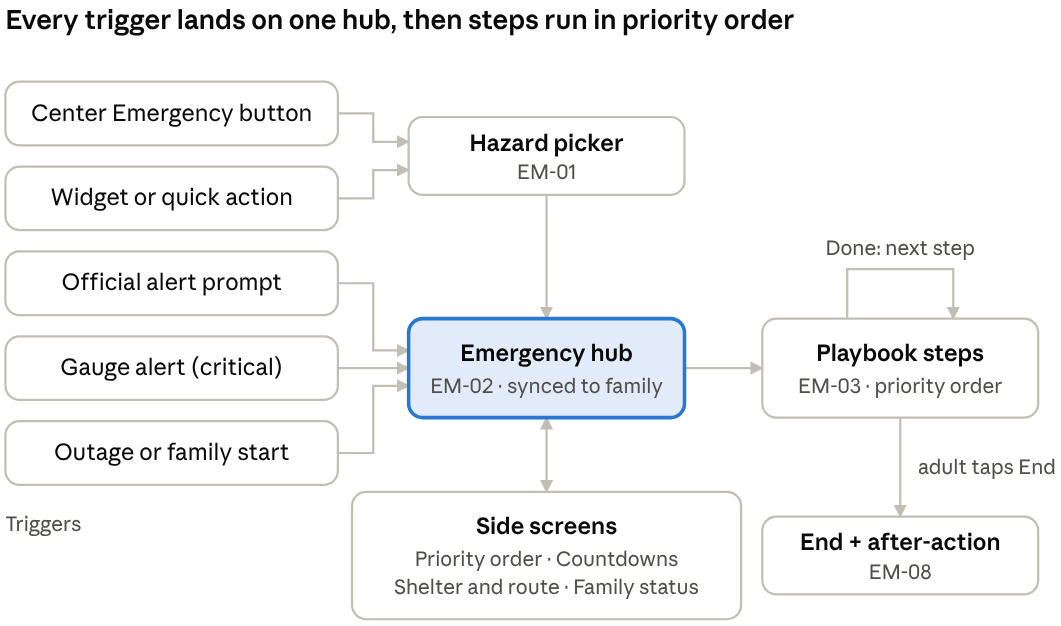
12. Emergency Mode

Emergency Mode is a dark, focused, step-by-step flow that walks the household through a specific hazard in priority order. It starts manually (center button) or from an official alert prompt, syncs across every household member's phone, and works fully offline.

12.1 Triggers

Trigger	Entry point	Hazard preselected?
Tap center Emergency button	`EM-01` Hazard picker	No (unless an alert is active; then that hazard is highlighted)
Alert full-screen prompt → "Start Emergency Mode"	`EM-02`	Yes
Home alert card → "Start Emergency Mode for [hazard]"	`EM-02`	Yes
Gauge alert (Critical)	`EM-02`	Flood
"Power is out" confirmed and user taps "Open outage steps"	`EM-02`	Power outage
Another member started it	Push: "Mom started Emergency Mode: Flood" → `EM-02` (joined)	Yes
Home-screen widget / lock-screen widget / quick action (long-press app icon → "Emergency")	`EM-01`	No

Starting Emergency Mode creates an `EmergencySession` synced to the household. Only one session per hazard can be active; starting a different hazard while one is active asks "Switch to Fire or run both?" (running both shows a hazard switcher at the top).



Emergency Mode flow · 5 triggers, 1 hub, step loop, end

Only the center button and widget go through the hazard picker; every alert-driven trigger opens the hub with the hazard already chosen.

12.2 EM-01 Hazard picker

- Full dark screen. Title: "What's happening?"
- Six large tiles in a 2×3 grid (each 50% width, 140 pt tall), with hazard icon and color: Flood · Winter storm · Power outage · Fire · Chemical/hazmat · Hurricane / high wind.
- If an alert is active, its tile has a glowing border and label "Alert active".
- Below the grid: a full-width red bar "Call 911" (always present), and a smaller "Something else" option that opens general steps (stay calm, check on everyone, call 911 if anyone is in danger, open the guide library).
- Tapping a tile goes to `EM-02` and creates the session.

12.3 EM-02 Emergency home (the hub)

This is the screen everyone sees during the emergency. Layout:

1. **Top bar:** hazard color stripe; hazard name and icon; elapsed time since start ("Flood · 12 min"); "Minimize" (returns to normal app with Return button) and "End" (top right, requires confirmation).

2. **Call 911 bar:** thin red bar, always visible, one tap to dial with confirmation sheet ("Call 911?").
3. **Alert strip** (if alert-triggered): event, expiry, "Official instructions" link.
4. **Next step card** (largest element, ~40% of screen): the first incomplete step from the playbook, in 22–28 pt text, with a big "Done" button and a smaller "Skip". If the step is Critical, it uses the critical treatment (red bar, pulse). Tapping the card opens `EM-03` at that step.
5. **Priority order row:** "Who first": horizontal avatars in rank order, Critical ones in the red group, with a check mark overlay once their steps are done. Tap → `EM-04`.
6. **Family status row:** each account holder's status (safe, on my way, need help, no response) as small avatars with status color. Tap → `EM-07`.
7. **Countdowns row** (outage, winter, hurricane, or any time power is out): the two shortest countdowns. Tap → `EM-05`.
8. **Shelter/route button** (flood, fire, hazmat evacuation, hurricane): "Where to go" → `EM-06`.
9. **Quick tools row:** Flashlight (toggles phone light), "Power is out" (if not yet confirmed), Share location (opt-in), Lock-screen card (shows it full-screen for responders), Guides (offline, filtered to this hazard).

12.4 EM-03 Step-by-step playbook

One step per screen, swipe or tap Done to advance. Each `StepCard`:

- Step number and total ("Step 3 of 11") and a thin progress bar.
- Tier treatment if Critical/Urgent.
- **Instruction:** one imperative sentence, max ~10 words ("Wake Grandma and move her upstairs.").
- **Assignee chip:** who should do it (from Tasks; e.g., "Rohan"), editable by tapping ("Reassign").
- **Detail** (collapsed by default, "How" expander): 2–4 short bullets.
- **Optional embedded tool:** checklist (go-bag items), timer (e.g., "Shelter for at least 1 hour, or until told it's safe"), call button (doctor, utility, medical supplier), countdown tile, or map snippet.
- **Buttons:** Done (64 pt, primary), Skip, and "Can't do this" (marks blocked and inserts the fallback step, e.g., "Call 911 for help moving Grandma").

- When a step is marked done on one phone, it shows done on all phones with "Done by Mom, 9:14 PM".
- Voice: a speaker button reads the step aloud in the user's language (on-device text-to-speech).

Step list view: a list icon in the top bar shows all steps with status (done, current, upcoming, skipped, blocked) so the user can jump around.

12.5 How playbooks are built

A playbook is generated at runtime from three layers:

1. **Hazard template:** fixed ordered steps for the hazard (12.8), each with an ID, tier, and conditions for inclusion.
2. **Subject steps:** for each Critical and Urgent subject in the priority order, the template inserts that subject's specific steps right after the hazard's "immediate safety" step, in rank order. For example, in an outage: "Check Grandma's oxygen battery" → "Switch Grandma to backup when below 30 min", then "Move Dad's insulin to the cooler".
3. **Household conditions:** steps included or removed based on the household (no basement → skip "Leave the basement"; no pets → skip pet steps; generator → add "Start generator outside, 20+ ft from windows").

Ordering rule: life-safety steps (get out, take shelter, call 911) always come first; then Critical subject steps in rank order; then Urgent subject steps; then general protective steps; then comfort and recovery.

12.6 EM-04 Priority order (emergency view)

Same data as PLAN-02 for the active hazard, but in Emergency styling: larger cards, "Act on these first" red group on top, and each card shows a progress indicator (e.g., "2 of 3 steps done") and the assigned helper's avatar. Tapping a card filters the playbook to that subject's steps. Order only; no reasons shown (long-press reveals them, as everywhere).

12.7 EM-05 Countdowns, EM-06 Shelter and route, EM-07 Family status, EM-08 End

EM-05 Countdowns: all running countdowns, sorted by time left, Critical first. Same tile as section 11.4, larger.

EM-06 Shelter and route:

- Map centered on home, showing open shelters (from the FEMA open shelter feed when online; cached list otherwise, labeled "may not be open"), hospitals, and meeting points.
- List below the map, sorted by distance, filters: Pet-friendly · Accessible (ADA) · Open now.
- Each row: name, distance, status (Open / Unknown / Closed) with "updated 20 min ago", pet-friendly paw, accessible icon, capacity if known.
- Row tap → detail sheet with address, phone, "Get directions" (hands off to Apple Maps / Google Maps with destination; offline: shows straight-line direction and the cached street map).
- For flood: a banner "Turn around, don't drown. Never drive through water." and a reminder that routes cannot see flooded roads.
- Meeting points appear first if the hazard is fire ("Meet at the mailbox") before shelters.

EM-07 Family status:

- One row per household member: avatar, name, status pill, last update time, location (if they chose to share), and for dependents, who is with them ("With Mom").
- My status buttons at the top: "I'm safe" (green), "On my way" (blue), "I need help" (red).
- "I need help" opens: "Call 911" button, and an optional short message to the household ("Stuck at work, roads flooded").
- "Ask everyone to check in" button sends a push to all account holders.
- Status changes sync instantly; members with no response for 15 minutes show amber "No response" with a call button.
- If offline, status updates queue and send when signal returns; the screen shows "Will send when connected". Also offers "Send by text message" which opens the SMS app pre-filled to household members' numbers ("I'm safe — at home."), since SMS often works when data does not.

EM-08 End Emergency Mode:

- "End" in the top bar → confirmation: "End Emergency Mode for everyone?" (Owner/adults) or "Leave Emergency Mode on this phone?" (others).
- Outcome picker: Safe at home · Evacuated · Went to a shelter · Other.
- Quick after-action screen: steps done/skipped/blocked summary, time taken, "What would you change?" free-text note, and suggestions generated by rules (e.g., "You skipped 'Grab go-bag' — it's missing 3 items" → link to supplies).

- For outages, opens the power-restored checklist (section 11.6).

12.8 Hazard playbooks (templates)

These are the default step templates. Steps in brackets are inserted per subject or per household condition. All steps link to an offline guide for detail. Content is based on standard public preparedness guidance (Ready.gov, CDC, NWS, American Red Cross) and must be reviewed by a qualified local reviewer before release.

Flood / flash flood

1. If water is rising around your home, get to higher ground now. Do not wait for instructions. (Critical)
2. Call 911 if anyone is trapped. (Critical)
3. [Get everyone out of the basement.] (Critical, only if basement)
4. [Per Critical/Urgent subject: "Help Grandma upstairs", "Carry Baby Aria".]
5. Grab the go-bag and critical meds. (Urgent; embedded checklist)
6. [Put Biscuit in the carrier.] (Important)
7. Turn off electricity at the main breaker only if you can do it without standing in water. (Important)
8. If told to evacuate, leave now. Never drive or walk through floodwater. (Critical when an evacuation alert is active)
9. Go to your meeting point or a shelter. (embedded map)
10. Check in with your family. (embedded status buttons)
11. Stay away from floodwater; it can hide hazards and carry contamination. (Standard)

Winter storm

1. Stay home and off the roads. (Important)
2. [Charge Grandma's oxygen batteries and spare batteries now.] (Critical, before outage)
3. [Confirm Dad has 7+ days of insulin.] (Urgent)
4. Charge phones and power banks. (Important)
5. Fill water containers in case pipes freeze. (Important)
6. Gather blankets and warm clothes in one room. (Important)
7. If power goes out: tap "Power is out" to start countdowns. (Urgent; embedded button) → switches to Power outage playbook steps.

8. Never use a grill, camp stove, or generator indoors or in a garage. (Critical when outage active)
9. [If dialysis at a center: call your center about schedule changes.] (Critical per WIN-C1)
10. Check on everyone every few hours. (Standard)

Power outage

1. Confirm the outage and note the time. (embedded "Power is out" button)
2. [Per life-sustaining device: "Check Grandma's oxygen — 4 h battery left".] (Critical; embedded countdown)
3. [If any life-sustaining device has under 8 h total backup: "Line up more power or a place to go" with options: generator, neighbor with power, call medical supplier, hospital as last resort, 911 if under 30 min.] (Critical)
4. Keep the fridge and freezer closed. (Urgent; embedded countdown)
5. [Move insulin to a cooler before the 4-hour mark — never directly on ice.] (Urgent → Critical under 30 min)
6. Report the outage to Dominion Energy. (Important; embedded call button and link)
7. [If freezing temps and no heat: gather everyone in one room, close doors, layer clothing, use blankets. Watch babies and older adults for cold signs.] (Critical per OUT-C3)
8. [If generator: run it outside, at least 20 ft from windows and doors.] (Critical)
9. Use flashlights, not candles. (Important)
10. Unplug sensitive electronics to protect them from surges. (Standard)
11. When power returns: tap "Power's back" and run the medication and food check. (embedded)

Fire (house fire)

1. Get out, stay out. Call 911 from outside. (Critical)
2. [Shut off Grandma's oxygen if safe, then get her out.] (Critical per FIR-C2)
3. [Per subject who can't self-evacuate, in rank order: "Get Baby Aria", "Help Grandma".] (Critical)
4. [Wake Rohan — he may not hear the alarm.] (Critical per FIR-C3; shown only if applicable)
5. Crawl low under smoke. Feel doors before opening; use another way out if hot. (Critical)
6. Go to the meeting point. (embedded meeting point map)

7. Count everyone. (embedded family checklist with every member and pet)
8. Never go back inside, even for pets. Tell firefighters who is missing. (Critical)
9. [Pets: tell firefighters where pets may be hiding.] (Important)

Chemical/hazmat (shelter in place by default)

1. Get inside now and stay inside. (Critical)
2. Close and lock all windows and doors. (Urgent)
3. Turn off heating, AC, and fans that pull in outside air. (Urgent)
4. Go to an interior room with few windows, above ground. (Urgent)
5. [Keep Grandma's oxygen and inhalers within reach.] (Critical per HAZ-C1)
6. Bring pets inside. (Important)
7. Listen for official updates and stay until told it's safe. (Urgent; embedded timer counting time sheltered)
8. If told to evacuate: follow official routes, leave immediately, keep windows closed and air on recirculate in the car. (Critical if evacuation alert)

Hurricane / high wind (inland)

1. Before it arrives: charge everything, fill water, gather meds for 7+ days. (Urgent)
2. [Check that every life-sustaining device has 72+ hours of backup, or a place to go.] (Critical)
3. Bring in or tie down outdoor furniture. (Standard)
4. During tornado warnings: go to the lowest floor, interior room, away from windows. [Mobile home: leave for a sturdy building before winds arrive.] (Critical)
5. Expect power outages: open the outage steps when power goes out. (embedded)
6. Watch for flooding: follow flood steps if water rises. (embedded gauge card)
7. After: stay away from downed power lines and report them. (Critical)

12.9 Emergency Mode rules

- Screen stays awake while Emergency Mode is open (user can disable).
- All text minimum 17 pt; Done buttons 64 pt tall; maximum two actions per screen besides Call 911.
- No ads, no upsells, no surveys, no non-emergency notifications while a session is active.

- Works entirely offline except shelters' live status, alerts, and family sync (all degrade gracefully with labels).
- Battery saver: dark palette always; map tiles load only on demand; location shared only when the user opts in.

13. Medication barcode scan

The user points the camera at the barcode on a medication box, Emercity extracts the National Drug Code (NDC), looks it up in the FDA's public NDC Directory through openFDA, and pre-fills the medication with name, strength, form, and the matching storage rule. It is deterministic (no AI or OCR guessing) and is the best single "wow" moment in a demo.

13.1 Flow

1. `HH-06` → "Scan barcode" (or from onboarding ONB-08, or the member profile).
2. `HH-07` camera screen opens.
3. On-device barcode detection reads the code (UPC-A, EAN-13, GS1 DataBar, GS1 DataMatrix).
4. The app extracts a candidate NDC (13.3).
5. Lookup: first the bundled offline NDC index (top refrigerated and critical medications, ~5–10 MB), then openFDA if online.
6. `HH-08` confirmation screen shows the match.
7. User picks the owner, confirms dose, opened/unopened, and quantity. Saved.

13.2 HH-07 Camera screen layout

- Full-screen camera preview with a rounded rectangle guide in the center ("Line up the barcode on the box").
- Torch toggle (top right), close X (top left).
- Below the guide: a hint that changes after 5 seconds without a read: "Try the barcode on the outer box, not the pharmacy sticker."
- Detected code flashes green with a light haptic, then the screen transitions automatically.
- Bottom: "Type it instead" link (opens search) and "Enter NDC number" (manual 10- or 11-digit entry with format helper).

13.3 Extracting the NDC

- **UPC-A / EAN-13 on U.S. drug packaging:** the NDC is embedded in the 12-digit UPC (number system 3). Strip the leading "3" and the check digit to get the 10-digit NDC.
- **GS1 DataMatrix / DataBar (newer packaging):** parse the GTIN (AI 01), then derive the NDC from the GTIN the same way. Also read expiration date (AI 17) and lot (AI 10) when present and pre-fill expiration.
- **10-digit NDC ambiguity:** a 10-digit NDC can be in 4-4-2, 5-3-2, or 5-4-1 format without dashes. The app tries all three hyphenations against the directory and keeps the one that matches a listed package.
- **Pharmacy labels:** many prescription bottle stickers carry the pharmacy's internal barcode, not an NDC. If no NDC can be derived, show "This looks like a pharmacy barcode. Scan the manufacturer's box, or type the medicine name."

13.4 openFDA lookup

- Endpoint: the openFDA drug NDC endpoint (`api.fda.gov/drug/ndc.json`) searching by `product_ndc` or `packaging.package_ndc`.
- Fields used: brand_name, generic_name, labeler_name, dosage_form, route, active_ingredients (name + strength), packaging description, product_type.
- Rate limits: openFDA works without a key at up to 240 requests per minute and 1,000 per day per IP; a free API key raises the daily limit to 120,000 (openFDA usage notes). Requests go through the Emercity server with its key and a 30-day cache, so individual phones never hit the limit.
- openFDA data is public domain.
- Note shown in the confirmation: the directory lists what the manufacturer submitted; FDA does not verify each entry, so the user should confirm the name matches the box.

13.5 HH-08 Scan result confirmation

Layout:

1. Success check and "Found it".
2. Product card: brand name (large), generic name, strength ("100 units/mL"), form and route ("Injection, solution · subcutaneous"), labeler, NDC (small, copyable).
3. Storage result box:

- If a StorageRule matches the NDC: snowflake icon, "Needs refrigeration", and safe-time summary ("Unopened: keep in fridge. Opened: up to N days below 86°F"), with source chip.
- If no rule: "We don't have storage details for this product yet. Check the label." and fields to enter refrigeration yes/no and room-temperature days.

4. Critical flag: auto-set for known critical classes (insulin, anticonvulsants, anticoagulants, immunosuppressants, epinephrine auto-injectors, inhaled rescue meds); visible toggle.
5. Owner picker: avatars of members and pets.
6. Dose and frequency (free text with suggestions), quantity on hand, opened/unopened toggle with date.
7. Buttons: "Save" (primary), "Scan another" (saves and reopens camera), "Not the right medicine" (back to camera).

13.6 Failure states

Case	Message	Action
Barcode read, NDC not found online or offline	"We couldn't find this medicine in the FDA directory."	"Type it instead"
Offline and not in the bundled index	"No connection. We saved the barcode and will look it up when you're online."	Saves a placeholder medication marked "Pending lookup"
Camera permission denied	"Camera access is off."	"Open Settings" + "Type it instead"
Multiple products match	List of matches to choose from	User taps the right one
Product discontinued	Shows the product with a gray "No longer listed" note	Save allowed

14. Lock-screen card generator (PLAN-09)

The lock-screen card is a generated wallpaper image that puts the household's most critical information on the phone's lock screen, readable without unlocking: meeting point, key contacts, and the top critical needs. Each member can make their own version. It is the only output the user chose, and it works on every phone with no app open.

14.1 Why a wallpaper

- It is visible to the owner and to first responders without unlocking the phone.
- It works with a dead network, a locked phone, and no app running.
- It complements, not replaces, the phone's built-in Medical ID, because it covers the whole household (meeting point, family contacts, dependents' needs).

14.2 Content blocks

The user toggles blocks on or off; defaults depend on whose card it is.

Block	Default (adult)	Default (teen)	Content
Header	On	On	"IN CASE OF EMERGENCY" + household name
Meeting points	On	On	Near-home spot and out-of-neighborhood spot, one line each
Emergency contacts	On	On	Two contacts with phone numbers + out-of-area contact
Critical needs	On	On	Top three household Critical items as short phrases: "Grandma (82): oxygen 24/7", "Dad: insulin-dependent diabetes"
My medical info	Off	Off	Owner's own conditions, allergies, blood type (owner opts in; sensitive)
Pets	On	On	"Dog Biscuit, microchip 985…" (last digits only by default)
Address	Off	Off	Home address (off by default for privacy if the phone is lost)

Block	Default (adult)	Default (teen)	Content
QR code	On	On	Encodes a small offline text summary (not a web link) that any phone camera can read (14.4)

14.3 PLAN-09 builder layout

1. **Live preview** (top 60% of screen): phone-shaped frame showing the generated wallpaper, with a faint overlay of where the clock and notifications sit on the lock screen, so the user can see nothing important will be hidden.
2. **"Whose card" picker:** avatars of account holders (each can make their own).
3. **Block toggles:** list of the blocks above with switches and drag handles to reorder.
4. **Style:** three themes: Dark (default, best legibility), Light, High contrast (yellow on black). Plus optional background tint by hazard season.
5. **Text size:** Standard / Large.
6. **Language:** card language (defaults to the owner's language; bilingual option prints key lines in two languages).
7. **Buttons:** "Save to Photos" (primary) with a one-time how-to sheet showing how to set it as the lock screen on iOS and Android; "Share" (sends the image); "Print" (prints a wallet-size version).

14.4 Layout rules for the generated image

- Rendered at the device's native resolution (e.g., 1290 × 2796 for large iPhones; Android uses the display size).
- Safe zones: nothing in the top 30% (clock and date) or the bottom 12% (flashlight/camera shortcuts and home indicator). Content lives in the middle band.
- Minimum text size equivalent to 16 pt on the lock screen; contact numbers 20 pt bold.
- Critical needs block uses a red left bar to echo the in-app Critical style.
- QR code (if on) is at least 25 mm wide on screen and encodes plain text like: "EMERCITY HOUSEHOLD · Meet: mailbox / Ashburn library · Contacts: Mom 703-555-0101, Uncle (out of area) 404-555-0199 · Critical: Grandma oxygen 24/7; Dad insulin". Plain text means responders can read it with any camera app, no internet needed.

14.5 Keeping it current

- When a block's underlying data changes (new contact, new Critical condition, new meeting point), the app shows a Home task: "Your lock-screen card is out of date" with "Update" (regenerates and prompts to save again). The app cannot set wallpapers automatically on iOS, so the user always confirms.
- Android: with permission, the app can set the lock-screen wallpaper directly via the system wallpaper API; the user chooses "Set automatically" or "Save only".
- In Emergency Mode, the quick tool "Lock-screen card" shows the card full-screen at max brightness for showing to a responder.

15. Practice drills

Drills let the household rehearse a hazard playbook with a timer, so the priority order and each person's role are practiced before a real emergency. They reuse Emergency Mode's flow in a clearly labeled practice mode and log results.

15.1 PLAN-06 Drills home

1. **Header:** "Practice makes it automatic" and the date of the last drill ("Last drill: fire, Aug 12 · 3 min 40 s").
2. **Recommended drill card:** chosen by rules: the top-risk hazard if never drilled; otherwise the hazard drilled longest ago; fire at least every 6 months (house fire applies to everyone).
3. **Drill types** (cards):
   - **Fire escape drill:** get everyone to the meeting point; measures time to all-out.
   - **Shelter-in-place drill (hazmat/tornado):** get everyone into the interior room with doors and windows closed.
   - **Flood move-up drill:** move everyone and the go-bag to the highest floor or out to the car.
   - **Outage drill:** walk through the outage steps; practice moving meds to the cooler and switching device batteries (no real power cut needed).
   - **Evacuation drill:** go-bag, pets in carriers, everyone in the car; measures time to drive-away-ready.
4. **History list:** past drills with date, type, time, participants, and a small trend sparkline per type.

5. **Schedule a drill:** date/time picker; invites household members via push and adds a calendar event (optional).

15.2 PLAN-07 Drill in progress

- Bright yellow "PRACTICE" banner across the top on every screen, so nobody mistakes a drill for a real emergency. Practice sessions never send real alerts or call 911; the Call 911 bar becomes "Practice: would call 911".
- Large elapsed timer (mm:ss) at the top.
- The same step cards as Emergency Mode, generated from the same playbook and priority order, with assignees.
- Each household member with the app sees the drill on their phone and taps "I'm at the meeting point" (or "I'm in the safe room") when done.
- Dependent tracking: the drill leader checks off dependents and pets as they arrive.
- "Stop drill" button (bottom).

15.3 PLAN-08 Drill results

- Total time and comparison to last time ("38 seconds faster").
- Per-person arrival times as a horizontal bar list, in priority order. Critical subjects' arrival time is highlighted: "Grandma reached the meeting point at 2:55".
- Steps skipped or marked "Can't do this", each with a suggestion (e.g., "Walker doesn't fit down the back stairs → set the front door as Grandma's exit").
- Rule-based tips: if a Critical subject arrived last, suggest assigning a dedicated helper; if total time exceeds a target (fire: 3 minutes is a common practice goal), suggest clearing exits or practicing again.
- Buttons: "Save", "Practice again", "Edit roles" (opens Tasks).
- Completing a drill updates the readiness score (Drills category).

15.4 Roles and tasks (shared with Emergency Mode)

- On PLAN-02 or in drill results, "Edit roles" opens a per-hazard assignment screen: each Critical/Urgent subject gets a helper (dropdown of capable members: adults and teens who can lift/drive as needed), and general tasks (grab go-bag, get pet carrier, turn off gas) get owners.
- The engine warns if one person is assigned to two Critical subjects in the same hazard ("Mom can't carry Aria and help Grandma at the same time").

- Roles appear as assignee chips on step cards in drills and real emergencies.

16. Map tab: shelters, resources, meeting points

The Map tab shows the household's surroundings through an emergency lens: risk layers, open shelters, hospitals and fire stations, river gauges, and the family's meeting points, all centered on home and available offline within 10 miles.

16.1 MAP-01 Map home layout

1. **Full-screen map** (vector tiles), centered on the household pin (a house icon in brand navy). User location shown as a blue dot when permitted.
2. **Search bar** (top, floating): search places and addresses.
3. **Layer button** (top right, floating) opens a bottom sheet with toggles:
   - Risk: Flood zones (FEMA) · County floodplain · Rail lines · Hazmat facilities
   - Places: Shelters · Hospitals and urgent care · Fire stations · Pharmacies · Cooling/warming centers
   - Live: Active alert areas · River gauges · Outage areas (when available)
   - Mine: Meeting points · Home
4. **Filter chips** (horizontal row under search): Open now · Pet-friendly · Accessible.
5. **Bottom sheet (collapsed):** "Nearby" list with the three closest places for the active layers; drag up for the full list.
6. **Recenter button** (bottom right, above the sheet).
7. **Legend button** explaining colors for flood zones and gauge categories.
8. Offline: map shows cached tiles within the 10-mile radius; beyond that, a gray area with "Not saved for offline".

Marker styles:

Layer	Marker	Color
Shelter (open)	House with heart	`status.safe` green
Shelter (status unknown)	House with heart, hollow	`status.stale` gray
Hospital / urgent care	Cross	Red

Layer	Marker	Color
Fire station	Helmet	Orange
Pharmacy	Pill	Teal
River gauge	Wave in circle	Colored by flood category
Meeting point	Flag	Brand navy, numbered 1 (near home), 2 (neighborhood), 3 (out of area)
Hazmat facility	Hazard diamond	Olive

16.2 MAP-02 Place detail sheet

- Name, type icon, status pill with "updated X min ago" and source (e.g., "FEMA shelter feed").
- Address (copy), phone (call), distance and estimated drive time when online.
- Shelters: pet-friendly, ADA accessible, capacity and current population when reported, hours.
- Hospitals: emergency department yes/no.
- Fire stations: "This station responds first to your home" badge if it's the household's first-due station.
- Buttons: "Directions" (hands off to the system maps app), "Save as meeting point" (for public places), "Share".

16.3 Data sources for places

Layer	Source	Status shown
Open shelters	FEMA National Shelter System open-shelter service, which includes status, open/closed dates, capacity, and population fields (FEMA NSS service)	Open / Closed / Unknown with last update

Layer	Source	Status shown
Potential shelter facilities (pre-identified, not necessarily open)	National Shelter System facilities; archives warn the layer must not be used to judge whether a facility is operating during an emergency (DataLumos archive)	Always "May not be open — check official updates"
County facilities model	Fairfax County's emergency facilities layer, used as the demo model where Loudoun data is unavailable	Labeled "modeled data"
Hospitals and urgent care	Loudoun County GIS	Static
Fire stations and first-due areas	Loudoun County GIS	Static
Pharmacies	Map provider's places search	Hours when available

Shelters are never shown as "open" unless a live source reported them open within the last 6 hours.

16.4 MAP-04 Meeting point editor

Three meeting points, each with a purpose:

1. **Near home** (fire): a spot just outside, e.g., "the mailbox" or "the neighbor's oak tree".
2. **Neighborhood** (if you can't return home): e.g., the community center or library.
3. **Out of area** (regional evacuation): a relative's home or a landmark in another town.

Editor layout:

- Map with a draggable flag pin; search field; "Use current location".
- Name field and description ("By the big oak at the end of the driveway"), optional photo (helpful for kids and visitors).
- Accessibility check: the app flags if a member with mobility needs would have to cross a hazard (e.g., the near-home point is down a steep hill) only if the user marks the path as having stairs or slopes; it never guesses terrain.

- Saved points appear on the lock-screen card, in fire and flood playbooks, and on EM-06.

16.5 MAP-05 Risk profile on the map

From PLAN-03 "See on map": the map opens with flood zones, floodplain, rail, and hazmat layers on, the household pin highlighted, and a 0.5-mile ring around home. Tapping a zone shows its designation and source.

17. Supply tracker and seasonal nudges

The supply tracker generates a go-bag and home-kit list tailored to the household (people, ages, conditions, pets, risks), computes quantities, and tracks what the family has and what expires. Seasonal nudges remind the household to prepare before high-risk seasons.

17.1 PLAN-04 Supplies layout

1. **Segmented control:** Go-bag · Home kit · Car (car is a simple list in v1).
2. **Progress header:** ring showing percent complete for the selected kit, plus counts ("34 of 41 items · 3 expiring soon").
3. **Filter chips:** All · Missing · Expiring · Critical.
4. **Category sections** (collapsible): Water · Food · Medical · Baby · Pets · Documents · Light and power · Tools · Hygiene · Clothing and warmth · Comfort.
5. **Item rows:** checkbox (have it), item name, quantity "have / need" ("6 / 15 gal"), tier marker if it supports a Critical need, expiration date (amber if within 30 days, red if expired), owner avatar if person-specific ("Aria: formula").
6. **"+ Add item"** at the bottom of each category for custom items.
7. **"Check everything" button:** starts a guided walkthrough (one item per screen: "Do you still have 2 flashlights? Batteries working?") that updates lastCheckedDate.

17.2 How the list is generated

Rules map household facts to items and quantities. Default duration: home kit for 7 days, go-bag for 3 days. Example rules:

Trigger	Item	Quantity rule	Kit
Every person	Drinking water	1 gallon per person per day	Home 7 days, go-bag 3 days
Every pet	Pet water	Dog/cat: 0.5 gallon per day (default, editable)	Home, go-bag
Every person	Non-perishable food	3 days go-bag, 7 days home	Both
Infant on formula	Ready-to-feed formula	Per daily feeds entered × days	Both (Critical)
Infant	Diapers, wipes	8 diapers/day default × days	Both
Critical medication	Extra supply	At least 7 days on hand; go-bag holds 3 days	Both (Critical)
Refrigerated medication	Cooler + ice packs + thermometer	1 per household	Home (Critical if insulin)
Oxygen / life-sustaining device	Spare batteries, backup supply, charger cables	Per device backup plan	Home (Critical)
CPAP	Battery pack or DC adapter	1	Home (Urgent)
Electric heat or outage risk High	Blankets, warm layers, hand warmers	1 blanket per person + extra	Home
Winter	Ice melt, shovel	1 each	Home
Flood zone High or basement	Sandbags (or water-filled barriers), sump pump battery backup check	Item	Home
Every home	Smoke alarms (test), CO alarms (test), fire extinguisher	1 per level / check date	Home
Generator owned	Fuel (stored safely), CO alarm	Fuel for N hours entered	Home

Trigger	Item	Quantity rule	Kit
Every person	Flashlight + batteries	1 per person	Both
Every household	Battery or hand-crank radio (NOAA Weather Radio)	1	Home
Every household	Phone power banks	1 per account holder	Go-bag
Every household	Documents: IDs, insurance, med list, prescriptions copy	Waterproof pouch	Go-bag
Every pet	Carrier, leash, food, meds, vaccination record copy, photo	1 set per pet	Go-bag
Member with hearing loss	Spare hearing aid batteries, visual/vibrating alarm	1 set	Both
Member with low vision	Spare glasses, white cane	1	Go-bag
Hazmat risk Moderate+	Plastic sheeting + duct tape (for sealing a room if instructed)	1 roll each	Home

When household facts change, the list updates; items the user added manually are never removed automatically.

17.3 PLAN-05 Supply item detail

- Item name, category, which kit, quantity have/need with stepper, unit.
- Why it's on the list (one line, e.g., "Grandma's oxygen concentrator" or "1 gallon per person per day for 5 people").
- Expiration date (scan barcode for food/meds or set manually), reminder toggle (30 days before).
- Location note ("Basement shelf, blue bin").
- Last checked date and "Checked today" button.
- Link to a guide (e.g., "How to store water").

17.4 Seasonal nudges

Nudges are scheduled notifications plus a Home card, triggered by calendar and the risk profile:

When	Nudge	Condition
Mid-October	"Winter is coming: test heat backup, check blankets, refill critical meds for snow days."	Winter score ≥ 40
Early June	"Hurricane season started. Make sure device backups can last 72 hours."	Hurricane score ≥ 30 or any life-sustaining device
March and September	"Daylight saving changes: test smoke and CO alarms, swap batteries."	Everyone
Late spring	"Thunderstorm season: check your sump pump and flashlight batteries."	Flood or outage score ≥ 40
Every 6 months	"Time for a fire drill."	Everyone
30 days before any supply expires	"Aria's formula expires Nov 20."	Item tracked
Refill date − 5 days	"Refill Dad's insulin by Friday."	Medication with refill date
Monthly	"Charge Grandma's spare oxygen batteries."	Device with spare batteries

Nudges respect quiet hours and can be turned off per type in Settings.

18. Guide library and grounded Q&A

The guide library is an offline, searchable collection of short, sourced articles that explain each step and situation in plain language. A small cloud Q&A sits on top as a subtle extra: it only answers from the library and official source documents, always shows its sources, and is never used in Emergency Mode's main flow.

18.1 PLAN-10 Guide library layout

1. Search bar ("Search guides — works offline"). Full-text search runs on device.
2. "For your household" row: 4–6 guides matched to members' conditions and top risks (e.g., "Insulin during a power outage", "Oxygen users: outage backup", "Babies in cold homes").
3. Hazard sections: Flood · Winter · Outage · Fire · Hazmat · Hurricane, each a horizontal carousel of guide cards.
4. Topics: Medications · Medical devices · Pets · Kids · Older adults · Disabilities · Food and water · Generators and safety · After the emergency.
5. Small "Ask a question" button at the bottom (opens `PLAN-12`, online only).

18.2 PLAN-11 Guide article layout

- Title, estimated read time ("2 min"), hazard tags.
- **"Do this now" box** at the top: 3–5 numbered imperative steps (the same wording used in playbooks).
- Body: short sections with headings, each under 100 words, grade 6 reading level.
- Inline callouts with tier styling for critical warnings ("Never use a generator indoors").
- **Sources** at the bottom: named organizations with links (FDA, CDC, Ready.gov, NWS, American Red Cross, county pages) and "Reviewed [date] by [reviewer]".
- Buttons: Read aloud (text-to-speech), Change language, Share.
- Language: guides ship pre-translated (section 20); machine translation is never used for safety text.

18.3 Starter guide list (version 1)

Hazard/topic	Guide titles
Outage	Keep insulin and refrigerated meds safe · Oxygen and other medical devices in an outage · Food safety when the power is out · Using a generator safely · Staying warm without power · Staying cool without power
Winter	Before a winter storm · Frozen pipes · Driving in snow (and when not to) · Shoveling and heart safety

Hazard/topic	Guide titles
Flood	Turn around, don't drown · Flash floods near creeks · Basements and flooding · After a flood: what's safe
Fire	Make a fire escape plan · Smoke alarms and Deaf/hard-of-hearing members · Home oxygen and fire safety · Getting kids and pets out
Hazmat	Shelter in place step by step · How to seal a room · If you're in the car
Hurricane	Inland hurricane effects · 72 hours of backup power for medical devices · Tornado warning: where to go
People	Babies and toddlers in emergencies · Older adults · Dementia and emergencies · Autism and sensory needs · Wheelchair and mobility users · Pregnancy
Pets	Pets in evacuations · Pet-friendly shelters · Pet first aid basics
After	Power's back: what to check · Medications after an emergency · Getting help and recovery resources

18.4 PLAN-12 Ask (grounded cloud Q&A)

This is the only AI feature, deliberately subtle and limited.

- Entry: a small "Ask" button in the guide library and at the bottom of guide articles. It is **not** in the tab bar, not on Home, and not inside Emergency Mode's step flow.
- Requires connection. Offline, the button shows "Ask needs a connection. Try searching guides instead."
- Behavior: the server retrieves relevant passages from the guide library and a fixed set of official documents (FDA, CDC, Ready.gov, NWS, Red Cross), and the model answers using only those passages. If no passage covers the question, it answers "I don't have official guidance on that. Call your pharmacist, doctor, or 911 if it's urgent."
- Every answer shows numbered source chips linking to the passages used; tapping opens the guide or official page.

- Household context: the question can optionally include non-identifying facts ("user has insulin pens, opened") toggled by the user; names, addresses, and other personal data are never sent.
- Hard rules (enforced in the system prompt and by output checks): no dosing changes, no diagnoses, no instructions that contradict official guidance, always defer to 911 for life-threatening situations, answers under 120 words.
- Banner on the screen: "Answers come from official guides. For emergencies, call 911."
- Feedback: thumbs up/down on each answer; flagged answers are reviewed.

19. Shared household plan (accounts and cloud sync)

Every adult and teen in the household signs in to their own account and sees the same live plan; changes sync through the cloud within seconds and work offline with automatic merge when the connection returns. This is the "digital" sync model chosen for v1.

19.1 Account model

- One account per person (Apple, Google, or email sign-in).
- A user can belong to several households (e.g., their own home and an elderly parent's home they help with). The household switcher sits in the app bar.
- Each household has exactly one Owner (the creator, 18+), who can transfer ownership.

19.2 Joining

- Owner or any Adult opens `HH-10` and shares an invite (code, QR, or link).
- The invitee installs the app, signs in, enters/scans the code, and chooses which existing member they are ("I'm Rohan") or adds themselves.
- Joining links their account to that Member record (`linkedUserId`).
- The owner gets a push: "Rohan joined The Patel Household."

19.3 Roles and permissions

Action	Owner	Adult	Teen (13–17)
View the full plan, priorities, countdowns	Yes	Yes	Yes

Action	Owner	Adult	Teen (13–17)
View other members' sensitive health details	Yes	Yes	Summary only (conditions as icons, not notes or medication doses) by default; owner can grant full view
Edit members, meds, devices	Yes	Yes	Own profile only
Edit supplies, meeting points, contacts	Yes	Yes	Supplies check-off only
Start Emergency Mode	Yes	Yes	Yes
End Emergency Mode for everyone	Yes	Yes	No (can leave on own phone)
Start drills	Yes	Yes	Yes
Invite members	Yes	Yes	No
Remove members, change roles, delete household	Yes	No	No
Change address	Yes	Yes (owner notified)	No

19.4 Sync architecture

- Data model: each household is a set of records (section 4) stored in a cloud document database with real-time listeners.
- Each phone keeps a full encrypted local copy (offline-first; section 21).
- Writes go to the local database first, then to the cloud via a queue. The UI never waits on the network.
- Real-time listeners push other members' changes to every phone in the household in under 2 seconds when online.
- Push notifications wake phones for important changes (Emergency Mode started, step done, check-in, outage confirmed, new Critical condition added) even when the app is closed.

19.5 Conflict rules

Situation	Rule
Two people edit different fields of the same record offline	Field-level merge: both changes kept
Two people edit the same field offline	Last writer wins by server timestamp; the overwritten value is kept in history for 30 days and the other editor sees "Mom changed this while you were offline"
Step marked done on two phones	Done once; credit to the first timestamp
Countdown start times differ (two people confirm outage at different times)	Earliest start time wins (safer)
Check-in statuses	Latest status per member wins
One person deletes a record another is editing	Deletion is soft for 30 days; the editor sees "This was removed by Dad" with "Restore"
Critical condition removed by one member while another adds a medication for it	Both changes apply; removing a Critical condition always notifies all adults

19.6 Activity feed

- Settings → Household → Activity shows recent changes: "Mom added insulin for Dad · 2 h ago", "Rohan checked off flashlights · yesterday".
- Emergency sessions show a timeline of steps and check-ins.
- Sensitive values are not printed in the feed ("Dad's medications were updated").

19.7 Dependents without phones

- Children and others without accounts are full Members in the plan but cannot sign in.
- In Emergency Mode, any adult can set a dependent's status ("Aria — with Mom, safe").
- Older relatives who don't use smartphones are reached through their lock-screen-style printed card (print option on PLAN-09) and the SMS fallback for check-ins.

20. Multilingual support and accessibility

Emercity must work for every member of the household, including people who don't read English well and people with disabilities, because they are often the ones most at risk. Language and accessibility are built into every component, not added later.

20.1 Languages

- **Launch set:** English and Spanish fully translated (UI, playbooks, guides, lock-screen card). Additional languages are chosen from the most-spoken non-English languages in Loudoun County according to Census American Community Survey data (verify before release; likely candidates include Korean, Vietnamese, Chinese, Hindi, Urdu, Arabic, Telugu, and Amharic).
- **Per-person language:** each Member has a primary language. Each account holder's app displays in their language. Lock-screen cards and printed plans can be bilingual (household language + member's language).
- **Safety text is never machine-translated.** Playbook steps, guides, and countdown messages ship as human-translated strings reviewed by a fluent speaker. Untranslated items fall back to English with a note, never to machine translation.
- **Official alerts:** NWS text may be English only. The app offers "Translate" clearly labeled "Machine translation — the original is below," and shows the mapped hazard and the household's own playbook (which is professionally translated) above it.
- **Formatting:** dates, times, numbers, and units follow the device locale (°F default for U.S., °C optional in Settings).
- **Right-to-left:** Arabic, Urdu, and Persian layouts mirror correctly (icons with direction like arrows flip; timers and numbers do not).

20.2 Accessibility requirements

Area	Requirement
Screen readers	Full VoiceOver (iOS) and TalkBack (Android) support. Every control has a label; PriorityCards read "Rank 1, Grandma, critical"; custom action "Why this rank" exposes the hidden reason
Dynamic text	Supports the largest accessibility text sizes; layouts reflow to one column; nothing truncates in Emergency Mode

Area	Requirement
Contrast	WCAG AA minimum everywhere; a High Contrast theme (yellow/white on black) in Settings
Color independence	Every tier and status has an icon and text label, not just color
Touch targets	48 pt minimum, 64 pt for Emergency Mode primary buttons
Motion	Respects Reduce Motion; no essential information conveyed by animation
Haptics	Critical items and step completions have distinct haptics, helpful for low-vision users; can be disabled
Audio	Read-aloud on every step and guide using on-device text-to-speech in the member's language
Deaf/hard of hearing	No information conveyed only by sound; alert sounds always paired with visual banners and vibration; Emergency Mode supports flashing the camera light on new critical events (opt-in)
Cognitive accessibility	Simple Mode (20.3); one action per screen in Emergency Mode; consistent button placement; icons beside words
Motor	All gestures have tap alternatives (no swipe-only actions); supports Switch Control and voice control labels

20.3 Simple Mode

A setting (per account) for older adults, kids, or anyone who wants less on screen:

- Home shows only three big buttons: "I need help" (calls 911 or a chosen contact), "Emergency steps", "My card" (lock-screen card full-screen).
- Emergency Mode shows one step at a time with pictures, read aloud automatically.
- No editing of the household plan (view only).
- Larger text by default; extra confirmation on calls.

20.4 Kid-friendly elements

- Teens and kids viewing steps assigned to them see simpler wording and a picture for each step (from a pictogram set) when Simple Mode is on.
- Meeting point photos help young children recognize the spot.

21. Offline-first architecture

Everything a household needs during an emergency runs on the phone with no connection: the plan, priority engine, playbooks, countdowns, guides, lock-screen card, cached map, and drills. Network features (alerts, live shelters, gauges, sync, Q&A) degrade gracefully with clear labels.

21.1 What works offline

Feature	Offline?	Notes
View and edit household	Yes	Edits queue and sync later
Priority engine	Yes	Runs locally; uses last known weather/alerts, labeled with their age
Emergency Mode playbooks	Yes	Templates bundled
Countdowns	Yes	Local timers and scheduled local notifications
Guides	Yes	Bundled + updated package
Lock-screen card	Yes	Generated on device
Medication scan	Partly	Bundled index of common critical/refrigerated meds; others queue for lookup
Map	Partly	Cached tiles and places within 10 miles of home; live status unavailable
Alerts	No	Shows last checked time; never says "All clear" offline
River gauges	No	Shows last reading with age
Shelter status	No	Cached list labeled "may not be open"

Feature	Offline?	Notes
Family sync and check-ins	No (queued)	SMS fallback for check-ins
Grounded Q&A	No	Suggests guide search

21.2 Local storage

- Encrypted local database (SQLite with SQLCipher, or the platform-encrypted store of the chosen sync SDK) holding the full household and reference data.
- Encryption key kept in the iOS Keychain / Android Keystore.
- **Offline package** (downloaded after onboarding and refreshed weekly on Wi-Fi): guides and playbooks in the user's languages (~5 MB), storage rules (~1 MB), bundled NDC index (~5–10 MB), places within 10 miles (~1 MB), map tiles for a 10-mile radius at street zoom (~30–80 MB, user can choose smaller radius).
- Settings → Offline data shows the package date, size, and a "Update now" button.

21.3 Sync queue

- Every local write creates a queued operation with a timestamp and household ID.
- On reconnect, the queue flushes in order; server applies merge rules (section 19.5) and returns any conflicts for display.
- The `OfflineBar` shows "Offline · 3 changes waiting to sync" and, after reconnecting, "Synced" for 3 seconds.

21.4 Freshness labels

Every live value has a freshness window. Past it, the `SourceChip` turns gray with "stale" and the age.

Data	Fresh for
Alerts	10 minutes
River gauge reading	1 hour
Shelter status	6 hours
Weather temperature (for rules)	3 hours

Data	Fresh for
Outage area counts	1 hour
Risk profile layers	45 days

When a rule depends on stale data (e.g., temperature for OUT-C3), the engine uses the last value but never lets staleness lower a tier: if the last known temperature was freezing, the subject stays Critical until fresh data says otherwise.

21.5 Battery and data use

- Background polling is done server-side; the phone relies on push, so it doesn't poll in the background.
- Emergency Mode uses dark palette, disables nonessential animations, and avoids continuous GPS (location only on request or when sharing is on, sampled every 5 minutes).
- A "Low battery" banner appears in Emergency Mode below 20% with tips (lower brightness, close other apps) and offers to show the lock-screen card and step list as a single screenshot-able summary.

22. Settings, privacy, and security

Settings gives each user control over their account, household, notifications, language, accessibility, privacy, and offline data. Privacy and security rules apply across the whole app because Emercity stores medical information.

22.1 SET-01 Settings home (grouped list)

1. **Profile:** avatar, name, email → `SET-02`.
2. **Household:** name, address, members and roles, invite, activity feed, switch household, leave household.
3. **Notifications** → `SET-03`.
4. **Language** → `SET-04`.
5. **Accessibility** → `SET-05` (text size override, High Contrast theme, Simple Mode, Reduce Motion override, haptics, read-aloud speed, flash light on critical events).

6. **Emergency preferences:** default hazard on the Emergency button (auto = top risk), keep screen awake in Emergency Mode, auto-start countdowns on outage suggestion (off by default), temperature unit.
7. **Privacy and data** → `SET-06`.
8. **Offline data** → `SET-07`.
9. **Data sources and credits** → `SET-08`.
10. **About** → `SET-09` (version, disclaimer, rule set review info, terms, privacy policy, contact).
11. **Sign out.**

22.2 SET-03 Notifications

- Master toggle; per-category toggles (section 23 lists categories).
- Critical alerts (official warnings, life-sustaining device countdowns, Emergency Mode started) are on by default and, where the OS supports it and approval is granted, can break through Do Not Disturb.
- Quiet hours for non-urgent categories (nudges, supply reminders).
- "Send a test alert" button.

22.3 SET-06 Privacy and data

- Plain-language summary: what is stored, who can see it (only household account holders), what is never done (sold, used for ads, shared with insurers or employers).
- Location: current permission state, "Share location during emergencies" default (ask each time).
- Q&A: toggle "Include non-identifying household context in questions" (off by default).
- Analytics: opt-in crash reports and anonymous usage counts; health fields are never included.
- **Export my data:** generates a JSON + readable PDF of the household plan for the requesting user.
- **Delete my account:** removes the user's account and unlinks their Member record (the Member stays as a dependent unless the owner deletes it).
- **Delete household** (owner only): two-step confirmation, 30-day recovery window, then permanent deletion from the cloud.

22.4 Security requirements

Area	Requirement
Transport	TLS 1.2+ for all traffic; certificate pinning for the Emercity API
At rest (cloud)	Database encryption at rest; sensitive health fields additionally encrypted per household with keys managed by a key management service
At rest (device)	Encrypted local database; key in Keychain/Keystore
Access control	Server-side rules: a user can only read/write households they belong to; role checks on every write (section 19.3)
App lock	Optional Face ID / fingerprint / passcode to open the app; Emergency Mode and the lock-screen card remain reachable without unlock (configurable)
Push payloads	Only IDs and generic text ("New alert for your home"); never names, conditions, or medications
Logging	No health or personal data in logs or crash reports
Third parties	Geocoding requests send only the address being searched; openFDA requests send only the NDC; Q&A sends only the question and opted-in non-identifying context
Minors	Users under 13 cannot have accounts; teens' data visible to household adults
Account recovery	Standard provider recovery (Apple/Google) or email link

22.5 SET-08 Data sources and credits

A screen listing every data source with name, what it's used for, link, update frequency, and whether it is real local data, national data, or modeled/demo data. Example rows: Loudoun County GIS (flood zones, hospitals, fire stations), NOAA National Water

Prediction Service (river gauges), National Weather Service (alerts, forecasts), FEMA National Shelter System (shelters), DOE/ORNL EAGLE-I (outage history), FDA openFDA NDC Directory (medications), FDA/CDC guidance (storage rules), Fairfax County GIS (modeled facility layer).

22.6 SET-09 About and disclaimer

Disclaimer text (shown at onboarding acceptance and here):

> Emercity helps your household plan for emergencies using public data and established guidance. It is not a substitute for official instructions, emergency services, or medical advice. Always follow orders from local officials and call 911 in an emergency. Medication and device times are estimates; confirm with your pharmacist, doctor, or equipment supplier.

Also shown: "Priority rules reviewed by [name, role] on [date]" once a reviewer has signed off.

23. Notification catalogue

Every notification Emercity can send, with its category, priority, delivery method, and destination. Copy is shown in English; all strings are translated. Push payloads never include health details (section 22.4).

ID	Category	Trigger	Example copy	Priority	Delivery	Opens
N-ALERT-W	Official alerts	Warning affecting address	"Flash Flood Warning for your home until 9:45 PM."	Critical	Remote push	ALERT-PROMPT
N-ALERT-A	Official alerts	Watch/advisory affecting address	"Winter Storm Watch for your area starting Friday."	Normal	Remote push	HOME-02
N-ALERT-END	Official alerts	Warning expired or cancelled	"The flood warning for your home has ended."	Low	Remote push	HOME-01

ID	Category	Trigger	Example copy	Priority	Delivery	Opens
N-GAUGE-F	Gauges	Gauge forecast to reach action stage	"Nearby creek forecast to reach action stage tomorrow."	Normal	Remote push	MAP-03
N-GAUGE-C	Gauges	Gauge at/forecast minor flood + high-risk home	"Nearby creek is flooding. Open flood steps."	Critical	Remote push	EM-02 (Flood)
N-EM-START	Emergency	Another member started Emergency Mode	"Mom started Emergency Mode: Flood."	Critical	Remote push	EM-02
N-EM-STEP	Emergency	A Critical step was completed or blocked	"Rohan: Grandma is upstairs." / "Mom can't move Grandma — needs help."	High	Remote push	EM-03
N-EM-CHECKIN	Emergency	Check-in requested	"Dad asked everyone to check in."	High	Remote push	EM-07
N-EM-HELP	Emergency	A member tapped "I need help"	"Mom needs help."	Critical	Remote push	EM-07
N-EM-NORESP	Emergency	Member no response 15 min	"Rohan hasn't checked in for 15 minutes."	High	Remote push	EM-07
N-OUT-ASK	Outage	Charger + Wi-Fi heuristic	"Did your power go out?"	Normal	Local	HOME-01 (outage prompt)
N-OUT-AREA	Outage	Area outage spike	"Outages reported near you. Is your power out?"	Normal	Remote push	HOME-01
N-OUT-CONF	Outage	Member confirmed outage	"Dad marked the power as out. Countdowns started."	High	Remote push	HOME-03
N-CD-FRIDGE	Countdown	Fridge phase ends in 30 min	"Move refrigerated medicine to the cooler within 30 minutes."	Urgent	Local (scheduled)	HOME-03

ID	Category	Trigger	Example copy	Priority	Delivery	Opens
N-CD-DEV2	Countdown	Life-sustaining backup under 2 h	"About 2 hours of backup left for a medical device."	Critical	Local (scheduled)	EM-05
N-CD-DEV30	Countdown	Life-sustaining backup under 30 min	"Under 30 minutes of device backup. Call 911 if you have no other source."	Critical	Local (scheduled)	EM-05
N-CD-FOOD	Countdown	Freezer near limit	"Freezer food: about 4 hours left if closed."	Low	Local	HOME-03
N-PWR-BACK	Outage	Member marked power restored	"Power's back. Check medicines and food."	Normal	Remote push	Power-restored checklist
N-SUP-EXP	Supplies	Item expires in 30 days	"Baby formula in your go-bag expires Nov 20."	Low	Local	PLAN-05
N-MED-REFILL	Medications	Refill date − 5 days	"Time to refill a critical medication by Friday."	Normal	Local	HH-05
N-DEV-CHARGE	Devices	Monthly	"Charge spare medical device batteries."	Normal	Local	HH-09
N-SEASON	Nudges	Seasonal schedule	"Winter is coming: check heat backup and blankets."	Low	Local	PLAN-04
N-DRILL	Drills	Scheduled drill or 6 months since last	"Fire drill tonight at 7 PM."	Normal	Remote push / local	PLAN-06
N-CARD	Lock-screen card	Card data changed	"Your lock-screen card is out of date."	Low	Local	PLAN-09
N-HH-JOIN	Household	Member joined	"Rohan joined your household."	Low	Remote push	HH-01
N-HH-CRIT	Household	Critical condition added or removed	"A household member's critical needs changed."	Normal	Remote push	HH-02

ID	Category	Trigger	Example copy	Priority	Delivery	Opens
N-SYNC	System	Sync conflict	"A change you made offline was updated by Mom."	Low	In-app only	Activity feed

Priority mapping:

- **Critical:** time-sensitive, uses the OS's highest priority (iOS Critical Alerts entitlement if granted, or time-sensitive interruption level; Android high-importance channel with sound and heads-up). Never batched or delayed.
- **Urgent/High:** time-sensitive interruption level; heads-up on Android.
- **Normal:** standard notification.
- **Low:** delivered quietly; may be grouped into a daily summary.

Rate limits: no more than one Normal/Low notification per day per category outside emergencies; Critical notifications are never rate-limited.

24. Tech stack, APIs, and data sources

Recommended stack: React Native with Expo (TypeScript) for the cross-platform app, Firebase for auth, real-time sync, cloud functions, and push, and MapLibre for offline maps. Flutter with the same Firebase backend is an equally valid alternative; pick whichever the builder knows better.

24.1 App (client)

Concern	Choice	Notes
Framework	React Native + Expo (TypeScript)	One codebase for iOS and Android; Expo dev builds for native modules
Navigation	React Navigation (bottom tabs + native stacks + modal for Emergency Mode)	Deep links configured per section 3.3
State	Zustand or Redux Toolkit for UI state; database as the source of truth for household data	

Concern	Choice	Notes
Local database	Firestore offline persistence for synced records + encrypted SQLite (expo-sqlite with SQLCipher) for reference data and sensitive-field cache	Or WatermelonDB if not using Firestore
Encryption	Client-side encryption of sensitive health fields with a per-household key (stored in Keychain/Keystore, shared to members via the server's key service on join)	Section 22.4
Maps	MapLibre GL (react-native-maplibre) with offline tile packs	Vector tiles from a hosted provider
Barcode	react-native-vision-camera with an on-device barcode scanner plugin (UPC/EAN/DataMatrix)	Deterministic decoding
Text-to-speech	expo-speech (on-device voices)	Per-member language
Notifications	expo-notifications + Firebase Cloud Messaging / APNs	Local scheduled notifications for countdowns
Background tasks	expo-task-manager (minimal; server does polling)	
Charts	react-native-svg + victory-native for gauge hydrographs	
Image generation (lock-screen card)	react-native-view-shot rendering an off-screen React view at device resolution	
i18n	i18next with ICU message format; RTL support	
Testing	Jest for logic (priority engine, countdown math, NDC parsing), Detox or Maestro for end-to-end flows	Engine fixtures per section 8.9

24.2 Backend

Concern	Choice
Auth	Firebase Authentication (Apple, Google, email link/password)
Database	Cloud Firestore with security rules enforcing household membership and roles
Server logic	Cloud Functions (Node/TypeScript): alert polling and matching, gauge watching, openFDA proxy and cache, invite handling, push fan-out, Q&A
Scheduling	Cloud Scheduler: alert poll every 2 min, gauge poll every 15 min, offline package rebuild weekly
Push	Firebase Cloud Messaging (Android + iOS via APNs)
Storage	Cloud Storage for photos and documents (encrypted, household-scoped paths)
Geospatial	Point-in-polygon with Turf.js in functions; precomputed household zone codes for NWS matching
Q&A	One function: retrieve passages from an embedded index of the guide library and official documents, call a hosted LLM with a strict grounded prompt, validate output (sources present, length, banned content), return answer + sources

24.3 External APIs and datasets

Source	Endpoint / access	Used for	Auth / limits
NWS alerts	`https://api.weather.gov/alerts/active?point={lat},{lng}` (per household) or by zone	Alerts (section 10)	No key; must send a User-Agent identifying the app

Source	Endpoint / access	Used for	Auth / limits
NWS forecast	`https://api.weather.gov/points/{lat},{lng}` → forecast/hourly URLs	Outdoor temperature for engine rules	No key
FEMA IPAWS public feed	IPAWS-OPEN public alert feed (CAP)	Non-weather alerts	Public
NOAA NWPS	`https://api.water.noaa.gov/nwps/v1/gauges/{id}` and `/gauges/{id}/stageflow` (NWPS API flyer)	River gauges, flood stages, forecasts	Public
FEMA National Shelter System	`https://gis.fema.gov/arcgis/rest/services/NSS/OpenShelters/FeatureServer` (layers)	Open shelters	Public ArcGIS REST
Loudoun County GIS	`https://logis.loudoun.gov/gis/rest/services/COL/FEMAFlood/MapServer`; hospitals `.../COL/HospitalsUrgentCare/MapServer/0`; GeoHub datasets	Flood zones, hospitals, fire stations, rail	Public ArcGIS REST
Fairfax County GIS	Open geospatial data services (model layers) (Fairfax open data)	Modeled facility/shelter layer	Public
DOE/ORNL EAGLE-I	County-level outage history datasets at 15-minute intervals (EAGLE-I 2025)	Outage frequency in risk profile	Download (historical, not a live API)
Live outage area counts	No free public live API identified; use member reports plus, optionally, a licensed outage data provider; demo can use simulated counts labeled as such	Area outage prompt	Varies
openFDA NDC	`https://api.fda.gov/drug/ndc.json` (openFDA NDC)	Medication lookup	Free key raises daily limit
HHS emPOWER	Public REST service of electricity-dependent Medicare beneficiaries by county/ZIP (emPOWER REST)	Context only (about screen, "why this matters" stats); never per-person	Public
Geocoding	Mapbox, Google, or the U.S. Census Geocoder	Address → coordinates, county	Key (Census is free)
Map tiles	MapTiler, Mapbox, or Stadia vector tiles	Map tab and offline packs	Key

24.4 Cloud functions (list)

1. `pollAlerts` (every 2 min): fetch active alerts for watched zones, match to households, write Alert records, send N-ALERT pushes.
2. `pollGauges` (every 15 min): fetch followed gauges, store readings, evaluate gauge triggers, send N-GAUGE pushes.
3. `ndcLookup` (HTTPS callable): parse/normalize NDC, query openFDA, attach StorageRule, cache 30 days.
4. `computeRiskProfile` (on address change + monthly): query all risk layers, compute factors and hazard scores, write RiskProfile.
5. `createInvite` / `acceptInvite` (callable): generate codes, link accounts, share household encryption key.
6. `fanOutEmergency` (Firestore trigger on EmergencySession/CheckIn/step writes): send N-EM pushes to household members.
7. `buildOfflinePackage` (weekly): bundle guides, storage rules, places per region.
8. `askGrounded` (callable): grounded Q&A (section 18.4).
9. `deleteHousehold` (scheduled after 30-day window): permanent deletion.

24.5 Firestore structure (sketch)

```
users/{userId}
households/{householdId}
  members/{memberId}
  pets/{petId}
  medications/{medicationId}
  devices/{deviceId}
  contacts/{contactId}
  meetingPoints/{pointId}
  supplies/{itemId}
  tasks/{taskId}
  drills/{drillId}
  sessions/{sessionId}
    checkIns/{checkInId}
    stepEvents/{eventId}
  alerts/{alertId}
  riskProfile/current
  activity/{eventId}
```

```
reference/storageRules/{ruleId}
reference/guides/{guideId}
reference/playbooks/{hazard}
reference/places/{placeId}
```

Security rules: a user may read/write `households/{id}/**` only if `request.auth.uid` is in that household's member list; role checks gate deletes and admin actions; `reference/**` is read-only for all signed-in users.

25. Empty, loading, error, and edge-case states

Every screen must handle empty, loading, offline, and error states explicitly; in a safety app, a blank or misleading screen is a failure. This section lists the required states screen by screen and the cross-cutting edge cases.

25.1 Empty states

Screen	Empty condition	Illustration + copy	Action
HH-01 Household	Only the owner added	Single figure in a house: "Add everyone who lives here so Emercity can plan for them."	"Add people and pets"
Medications	None added	Pill bottle: "Add medicines that matter in an emergency, especially ones that need the fridge."	"Scan a box" / "Type it"
Devices	None	Plug: "Does anyone use medical equipment that needs power?"	"Add equipment" / "No one does"
PLAN-02 Priorities	No members beyond owner	"Your order will appear once you add your household."	"Add people"
PLAN-04 Supplies	Not generated yet	Box: "We'll build your list from your household."	"Build my list"
PLAN-06 Drills	No drills	Stopwatch: "Practice once and you'll move faster for real."	"Start a fire drill"

Screen	Empty condition	Illustration + copy	Action
MAP meeting points	None set	Flag: "Pick a spot everyone knows."	"Set meeting point"
HOME countdowns	No power-dependent items	"Nothing in your home needs a countdown. Food timers start when power goes out."	—
EM-07 Family status	No other account holders	"Invite your household so you can see who's safe."	"Invite"
Alert history	No alerts in a year	"No alerts for your address in the last year."	—

25.2 Loading states

- Skeleton cards (gray shimmering blocks matching the final layout) for any content that loads from the network; local content renders instantly.
- Never show a spinner without a label for more than 2 seconds: "Checking alerts for your address…", "Looking up this medicine…".
- Risk profile reveal uses a staged loader that shows each factor as it arrives.

25.3 Error states

Error	Where	Behavior
No network	Any network feature	`OfflineBar`; cached data with age; never "All clear"
API timeout	Alerts, gauges, shelters	Retry with backoff (2, 4, 8, 16 s); show last known with stale label; "Retry" button
Data source down	Any	Banner naming the source ("River gauge data is unavailable right now")
Sync failure	Writes	Changes stay queued; badge on Settings → Household; never silently lost

Error	Where	Behavior
Geocode fails	Onboarding	"We couldn't find that address. Drop a pin on the map instead."
Push token invalid	Background	Re-register silently; if notifications disabled, Home shows an "Alerts are off" card
Camera unavailable	Scan	Manual entry
Storage full	Offline package	"Not enough space for offline maps. Save a smaller area?" with radius options (2, 5, 10 mi)
App update required (rules changed)	Launch	Non-blocking banner; never block access to Emergency Mode

25.4 Edge cases

- **Two hazards at once** (e.g., winter storm plus outage, or flood plus outage): Emergency Mode supports multiple active hazards; the priority order merges by taking each subject's highest tier across active hazards, and the playbook interleaves life-safety steps from both first.
- **Member away from home during an emergency:** check-in statuses cover this; steps assigned to an absent member auto-suggest reassignment ("Dad is away. Reassign 'Get Grandma' to Mom?").
- **Traveling:** alerts follow the home address; with Always location on, the user can also get alerts for their current location (labeled "Where you are" vs "At home").
- **Multiple households:** Emergency Mode is per household; the hazard picker asks which household if the user belongs to more than one and alerts affect both.
- **Household with no adults signed in** (e.g., teen alone): Emergency Mode works fully; "End for everyone" requires an adult; teen sees an extra "Call a parent" button.
- **Device clock wrong:** countdowns use monotonic elapsed time from the start event, not wall-clock math, so a time-zone or clock change doesn't break them.
- **App killed during an outage:** countdown notifications are pre-scheduled locally, so they still fire; reopening the app restores state from the database.
- **Phone dies:** another member's phone holds the same plan; the lock-screen card and printed plan are the fallback.

- **Medication with no storage rule:** never guessed; user enters label values (section 11.2).
- **Address outside the U.S.:** app works for household planning and countdowns; risk layers, alerts, and shelters show "Not available in your area".
- **Very large households (10+ members):** priority lists virtualize; lock-screen card shows top three Critical items only, plus "+N more in the app".
- **Deceased or moved-out member:** "Archive member" keeps history but removes them from plans and cards.
- **Language missing for a guide:** English with a note "Not yet available in [language]"; never machine translation for safety text.

26. Feature priorities, build order, and handoff notes

Build the five headline features first and make them excellent; they carry the pitch. Everything else supports them. Scores use the weighting agreed earlier: Judge impact 35%, Technical depth 25%, Demo value 25%, Trust and accuracy 15%.

26.1 Feature priority table

#	Feature	Section	Score	Tier
1	Priority engine	8	92	Headline
2	Address risk profile	9	88	Headline
3	Live alerts + river gauge watch	10	88	Headline
4	Outage countdown	11	87	Headline
5	Personalized Emergency Mode	12	85	Headline
6	Medication barcode scan	13	88	Core (best demo moment)
7	Lock-screen card	14	80	Core
8	Offline-first	21	76	Core
9	Practice drills with roles	15	75	Core
10	Guided setup with quick-add chips	5	71	Required flow
11	Open shelters with freshness labels	16	70	Strong add
12	Multilingual (human-translated)	20	70	Strong add

#	Feature	Section	Score	Tier
13	Shared household plan (cloud sync)	19	68	Strong add
14	Resource map	16	64	Strong add
15	Supply tracker	17	63	Strong add
16	Grounded cloud Q&A (subtle)	18	60	Subtle extra
17	Meeting points	16	54	Table stakes
18	Seasonal nudges	17	51	Subtle extra

26.2 Build order

Each phase produces something that runs. Later phases depend on earlier ones.

1. **Foundation:** project setup, design system tokens and core components (section 2), navigation shell with tabs and Emergency button (section 3), local database and data model (section 4), Firebase auth.
2. **Household and setup:** onboarding with quick-add chips (section 5), Household tab (section 7), manual medication and device entry.
3. **Priority engine:** rules, scoring, fixtures and tests (section 8), PLAN-02 display with tier treatment, Home top-priorities strip.
4. **Outage countdown:** storage rules data, countdown math, tiles, local notifications, "Power is out" / "Power's back" flows (section 11).
5. **Emergency Mode:** hazard picker, hub, step cards, playbook generation from templates + priority order, End flow (section 12). Single-device first.
6. **Risk profile:** geocoding, Loudoun layers, hazard scores, reveal screen and PLAN-03 (section 9).
7. **Alerts and gauges:** cloud polling functions, push, alert prompt, gauge card and detail (section 10).
8. **Barcode scan:** camera, NDC parsing, openFDA proxy, confirmation screen (section 13).
9. **Lock-screen card:** builder and image generation (section 14).
10. **Shared plan:** invites, roles, real-time sync, multi-device Emergency Mode, family status board (section 19, section 12.7).
11. **Map:** MapLibre, layers, shelters, meeting points, offline tiles (section 16).
12. **Drills, supplies, nudges, guides** (sections 15, 17, 18).

13. **Offline hardening, accessibility pass, translations** (sections 20, 21).
14. **Grounded Q&A** (section 18.4), last, because it is the least essential.
15. **Review and polish:** rule set reviewed by a qualified local reviewer, data sources screen complete, every empty/error state from section 25 implemented.

26.3 Demo household (seed data)

Ship a built-in demo household so the app can be shown without real personal data. Use the worked example from section 8.6 (Grandma with oxygen, Dad with insulin, Mom, Baby Aria, teen Rohan, dog Biscuit) at a sample Leesburg, VA address in a mapped flood zone near a creek gauge, with a simulated winter-storm outage scenario and a simulated flash flood warning that can be triggered from a hidden demo menu. All simulated data is labeled "Demo" on screen.

26.4 Acceptance checklist (definition of done for v1)

- [ ] A new user completes onboarding and sees a correct priority order in under 5 minutes.
- [ ] Priority engine passes all fixture tests; Critical subjects always appear first and pulse.
- [ ] Turning on airplane mode: Emergency Mode, countdowns, guides, lock-screen card, and cached map all still work.
- [ ] A simulated flash flood warning produces a push, the full-screen prompt, and a flood playbook ordered by the household's priorities.
- [ ] Confirming an outage starts countdowns on every household member's phone with the same start time.
- [ ] Scanning an insulin box fills in the product and its storage rule with a cited source.
- [ ] Two members editing offline then reconnecting merge correctly per section 19.5.
- [ ] Every live value shows its source and age; stale data is labeled; the app never says "All clear" when it can't check.
- [ ] VoiceOver/TalkBack can complete onboarding and an entire Emergency Mode playbook.
- [ ] Lock-screen card renders correctly on at least one small and one large phone in both iOS and Android.
- [ ] Data sources screen lists every source and marks modeled/demo data.

26.5 Handoff prompt (paste into another chat)

Use this to hand the spec to another AI chat or a developer:

> You are building Emercity, a cross-platform (React Native + Expo, TypeScript, Firebase) household emergency app. The attached specification is the source of truth. Build in the phase order in section 26.2, one phase at a time. For each phase: list the files you will create, write complete code (no placeholders), include unit tests for any logic (priority engine, countdown math, NDC parsing), and stop at the end of the phase for review. Never use AI for any emergency decision; the priority engine and playbooks are rule-based exactly as specified in sections 8 and 12. Respect the design tokens in section 2, the data model in section 4, and the privacy rules in section 22. Use the demo household in section 26.3 as seed data. Start with phase 1.

26.6 Open questions

- Who will review the priority rules and playbooks (local emergency manager, nurse, or CERT instructor), and by what date?
- Which additional languages beyond English and Spanish, based on Loudoun Census data?
- Is a licensed live outage data feed worth the cost, or do member reports plus demo data suffice for v1?
- Future features to scope later: car plans, away-from-home plans, school pickup.

Sources

- Loudoun County FEMA Flood Hazard Zones map service
- Loudoun Floodplain dataset (data.gov)
- Loudoun Hospitals and Urgent Care (data.gov)
- Loudoun first dues and station locations
- Loudoun County winter storm guidance, Jan 2026
- Fairfax County open geospatial data
- NOAA NWPS API flyer
- FEMA National Shelter System service layers
- FEMA Open Shelters feature service
- HIFLD National Shelter System facilities archive (DataLumos)
- DOE/ORNL EAGLE-I power outage data 2025
- openFDA NDC Directory
- FDA insulin storage and switching in an emergency

- Vanderbilt poison center: refrigerated medication stability
- HHS emPOWER public REST service